

kohlhase-omdoc: image evidence

Image regions

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0001	19	—	—	—	$\section[\text{\TeX}]{\text{The Joy of \text{\TeX}}\index{tex@\text{\TeX}}\label{sec:TeX}}$
kohlhase-omdoc_DIA_0002	19	—	—	—	$\{\text{\bf proof}\}:\dots\hfill\Box$
kohlhase-omdoc_DIA_0003	22	—	$\begin{lstlisting}[mathescape=true] <omtext xml:id=" foo" xmlns="http://www.mathweb.org/omdoc" xmlns:om="http://www.openmath.org/OpenMath" > <CMP xml:lang='en'> The number <om:OMOBJ><om:OMS cd="nums1" name="pi"/><om:OMOBJ> is irrational. </CMP> </omtext> \end{lstlisting}$	(not rendered)	$\begin{array}{c} \text{omtext} \\ \swarrow \quad \searrow \\ \begin{array}{ c c } \hline \text{xml:id} & \text{foo} \\ \hline \end{array} \quad \begin{array}{ c } \hline \text{CMP} \\ \hline \end{array} \\ \swarrow \quad \downarrow \quad \searrow \\ \begin{array}{ c } \hline \text{om:OMOBJ} \\ \hline \end{array} \\ \swarrow \quad \downarrow \quad \searrow \\ \begin{array}{ c c c } \hline \text{om:OMS} & & \\ \hline \end{array} \quad \begin{array}{ c c } \hline \text{cd} & \text{nums1} \\ \hline \end{array} \\ \swarrow \quad \downarrow \quad \searrow \\ \begin{array}{ c } \hline \text{text} \\ \hline \end{array} \quad \begin{array}{ c c } \hline \text{name} & \text{pi} \\ \hline \end{array} \quad \begin{array}{ c } \hline \text{text} \\ \hline \end{array} \\ \swarrow \quad \downarrow \quad \searrow \\ \begin{array}{ c } \hline \text{The number} \\ \hline \end{array} \quad \begin{array}{ c c } \hline & \\ \hline \end{array} \quad \begin{array}{ c } \hline \text{is irrational.} \\ \hline \end{array} \end{array}$
kohlhase-omdoc_DIA_0004	24	—	—	—	$\begin{array}{l} <?xml\ version="1.0"?> \\ <!DOCTYPE\ omdoc\ PUBLIC\ "</math>$
kohlhase-omdoc_DIA_0005	30	—	—	—	$\begin{array}{c} \begin{array}{ c } \hline 3 \\ \hline \end{array} \\ \begin{array}{ c } \hline (x+2) \\ \hline \end{array} \\ \swarrow \quad \searrow \\ \begin{array}{ c } \hline 3 \\ \hline \end{array} \quad \begin{array}{ c c c } \hline (& x+2 &) \\ \hline \end{array} \\ \swarrow \quad \downarrow \quad \searrow \\ \begin{array}{ c } \hline x \\ \hline \end{array} \quad \begin{array}{ c } \hline + \\ \hline \end{array} \quad \begin{array}{ c } \hline 2 \\ \hline \end{array} \end{array}$
kohlhase-omdoc_DIA_0006	30	—	—	—	$\begin{array}{c} @(\cdot,\cdot) \\ \swarrow \quad \downarrow \quad \searrow \\ \text{division} \quad 3 \quad @(\cdot,\cdot) \\ \swarrow \quad \downarrow \quad \searrow \\ \text{addition} \quad x \quad 2 \end{array}$

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kohlhase-omdoc_DIA_0007	31	—	—	—	
kohlhase-omdoc_DIA_0008	32	—	—	—	<pre> <OMOBJ xmlns="http://www.openmath.org/OpenMath"> <OMBIND cdbase="http://www.openmath.org/cd"> <OMS cd="quant1" name="forall"/> <OMBVAR><OMV name="a"/><OMV name="b"/></OMBVAR> <OMA><OMS cd="relation" name="eq"/> <OMA><OMS cd="arith1" name="plus"/> <OMV name="a"/> <OMV name="b"/> </OMA> <OMA><OMS cd="arith1" name="plus"/> <OMV name="b"/> <OMV name="a"/> </OMA> </OMBIND> </OMOBJ> </pre>
kohlhase-omdoc_DIA_0009	33	—	—	—	<pre> <CDDefinition> <Name>plus</Name> <CDDescription> The symbol representing an n-ary commutative function plus. </CDDescription> <CMP> for all a,b a + b = b + a </CMP> <FMP> ∀a, b. a + b = b + a </FMP> </CDDefinition> </pre>
kohlhase-omdoc_DIA_0010	41	—	—	—	
kohlhase-omdoc_DIA_0011	42	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0012	50	—	—	—	<pre> <?xml version="1.0" encoding="utf-8"?> <!DOCTYPE omdoc PUBLIC "-//OMDoc//DTD OMDoc Basic V1.2//EN" "http://www.mathweb.org/omdoc/dtd/omdoc-basic.dtd" []> <omdoc xml:id="algebra1.omdoc" version="1.2" modules="@basic" xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:cc="http://creativecommons.org/ns" xmlns="http://www.mathweb.org/omdoc" > <metadata> <dc:title>Laws of Composition</dc:title> <dc:creator role="trl">Michael Kohlhase</dc:creator> <dc:date action="created">2002-01-03T07:03:00</dc:date> <dc:date action="updated">2002-11-23T18:17:00</dc:date> <dc:description> A first migration step for a fragment of Bourbaki's Algebra </dc:description> <dc:source> Nicolas Bourbaki, Algebra, Springer Verlag 1989, ISBN 0-387-19373-1 </dc:source> <dc:type>Text</dc:type> <dc:format>application/omdoc+xml</dc:format> <dc:rights>Copyright (c) 2005 Michael Kohlhase</dc:rights> <cc:license> <cc:permissions reproduction="permitted" distribution="permitted" derivative_works="permitted"/> <cc:prohibitions commercial_use="permitted"/> <cc:requirements notice="required" copyleft="required" attribution="required"/> </cc:license> </metadata> <omtext xml:id="all"> <CMP xml:lang="en"> {\sc Definition 1.} Let E be a set. A mapping $E \times E$ is called a law of ... mappings from E into E ($\{\emph{Set Theory}\}$), II, §5, no. 2). </CMP> </omtext> </omdoc> </pre>
kohlhase-omdoc_DIA_0013	52	—	—	—	<pre> <omtext xml:id="magma.def" type="definition"> <CMP>Let <legacy format="TeX">E</legacy> be a set ... called a magma.</CMP> </omtext> <omtext xml:id="t1"> <CMP>The composition of <legacy format="TeX">x</legacy> ... multiplicatively.</CMP> </omtext> <omtext xml:id="t2"> <CMP>In the general ... composition.</CMP> </omtext> <omtext xml:id="t3"> <CMP>By an abuse ... on <legacy format="TeX">E</legacy></CMP> </omtext> <omgroup xml:id="magma-ex" type="enumeration"> <metadata><dc:title>Examples</dc:title></metadata> <omtext type="example" xml:id="e1.magma"> </pre>
kohlhase-omdoc_DIA_0014	53	—	—	—	<pre> <CMP> The mappings <legacy format="TeX">(X, Y)</legacy> ... subsets of a set <legacy format="TeX">E</legacy>. </CMP> </omtext> <omtext type="example" xml:id="e2.magma"> <CMP> On the set <legacy format="TeX">$\mathbf{N}$</legacy> ... III, §3, no. 4). </CMP> </omtext> <omtext type="example" xml:id="e3.magma"> <CMP> Let <legacy format="TeX">E</legacy> be a set; ... II, §5, no. 2). </CMP> </omtext> </omgroup> </pre>

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kohlhase-omdoc_DIA_0015	54	—	—	—	<pre> <!DOCTYPE omdoc PUBLIC "-//OMDoc//DTD OMDoc CD V1.2//EN" "http://www.mathweb.org/omdoc/dtd/omdoc-cd.dtd" [<!ENTITY % om.prefixed "INCLUDE">]> <theory xml:id="magmas"> <imports from="background.omdoc#products"/> <imports from="http://mbc.mathweb.org/omstd/relation1.omdoc#relation1"/> <symbol name="magma"> <metadata><dc:description>Magma</dc:description></metadata> </symbol> <symbol name="law_of_composition"/> <definition xml:id="magma.def" for="#magma #law_of_composition"> <CMP> Let <om:OMOBJ><om:OMV name="E"/></om:OMOBJ> be a set. A mapping of <om:OMOBJ> <om:OMA><om:OMS cd="products" name="Cartesian-product"/> <om:OMV name="E"/><om:OMV name="E"/> </om:OMA> </om:OMOBJ> is called a <term cd="magmas" name="magma" role="definiens">law of composition</term> on <om:OMOBJ><om:OMV name="E"/></om:OMOBJ>. The value <om:OMOBJ> <om:OMA><om:OMV name="f"/> <om:OMV name="x"/><om:OMV name="y"/> </om:OMA> </om:OMOBJ> of <om:OMOBJ><om:OMV name="f"/></om:OMOBJ> for an ordered pair <om:OMOBJ> <om:OMA><om:OMS cd="sets" name="in"/> <om:OMA><om:OMS cd="products" name="pair"/> <om:OMV name="x"/><om:OMV name="y"/> </om:OMA> <om:OMA><om:OMS cd="products" name="Cartesian-product"/> <om:OMV name="E"/><om:OMV name="E"/> </om:OMA> </om:OMA> </om:OMOBJ> is called the <term cd="magmas" name="law_of_composition" role="definiens-applied">composition</term> of <om:OMOBJ><om:OMV name="x"/></om:OMOBJ> and <om:OMOBJ><om:OMV name="y"/></om:OMOBJ> under this law. A set with a law of composition is called a <term cd="magmas" name="magma" role="definiens">magma</term>. </CMP> </definition> ... </theory> ... </pre>
kohlhase-omdoc_DIA_0016	56	—	—	—	<pre> <theory xml:id="magmas-examples"> <metadata><dc:title>Examples</dc:title></metadata> <imports from="http://mbc.mathweb.org/omstd/fns1.omdoc##fns1"/> <imports from="background.omdoc#nat"/> <imports from="background.omdoc#functions"/> <imports from="#magmas"/> <omgroup xml:id="magma-ex" type="enumeration"> <metadata><dc:title>Examples</dc:title></metadata> <example xml:id="e1.magma" for="#law_of_composition" type="for"> <CMP>The mappings <om:OMOBJ> <om:OMBIND><om:OMS cd="fns1" name="lambda"/> <om:OMBVAR> <om:OMV name="X"/><om:OMV name="Y"/> </om:OMBVAR> </om:OMOBJ> </example> </omgroup> </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0017	57	—	—	—	<pre> </om:OMBVAR> <om:OMA><om:OMS cd="functions" name="pattern-defined"/> <om:OMA><om:OMS cd="products" name="pair"/> <om:OMV name="X"/> <om:OMV name="Y"/> </om:OMA> <om:OMA><om:OMS cd="sets" name="union"/> <om:OMV name="X"/> <om:OMV name="Y"/> </om:OMA> </om:OMA> </om:OMBIND> </om:OMOBJ> and <om:OMOBJ> <om:OMBIND><om:OMS cd="fns1" name="lambda"/> <om:OMBVAR> <om:OMV name="X"/><om:OMV name="Y"/> </om:OMBVAR> <om:OMA><om:OMS cd="functions" name="pattern-defined"/> <om:OMA><om:OMS cd="products" name="pair"/> <om:OMV name="X"/> <om:OMV name="Y"/> </om:OMA> <om:OMA><om:OMS cd="sets" name="intersection"/> <om:OMV name="X"/> <om:OMV name="Y"/> </om:OMA> </om:OMBIND> </om:OMOBJ> are <term cd="magmas" name="law_of_composition">laws of composition</term> on the set of subsets of a set <om:OMOBJ><om:OMS cd="magmas" name="E"/></om:OMOBJ>. </CMP> </example> <example xml:id="e2.magma" for="#law_of_composition" type="for"> <CMP> On the set <om:OMOBJ><om:OMS cd="nat" name="Nat"/></om:OMOBJ> of <term cd="nats" name="nats">natural numbers</term>, <term cd="nats" name="plus">addition</term>, <term cd="nats" name="times">multiplication</term>, and <term cd="nats" name="power">exponentiation</term> are ... </CMP> </example> </omgroup> </theory> </pre>
kohlhase-omdoc_DIA_0018	58	—	—	—	<pre> <theory xml:id="magmas"> <imports xml:id="imp3" from="background.omdoc#products"/> <imports from="http://mbc.mathweb.org/omstd/relation1.omdoc#relation1"/> ... <theory xml:id="magmas-examples" from="http://www.mathweb.org/omdoc/examples/omstd/fns1.omdoc#fns1"/> <imports xml:id="imp5" from="background.omdoc#nat"/> <imports xml:id="imp6" from="background.omdoc#functions"/> ... </theory> ... </theory> </pre>
kohlhase-omdoc_DIA_0019	58	—	—	—	<pre> <symbol name="law_of_composition"> <metadata><dc:description>A law of composition on a set.</dc:description></metadata> </symbol> <definition xml:id="magma.def" for="#law_of_composition" type="simple"> <CMP> Let <om:OMOBJ><om:OMV name="E"/></om:OMOBJ> be a set. A mapping of <om:OMOBJ><om:OMR href="#comp.1"/></om:OMOBJ> is called a <term cd="magmas" name="law_of_composition" role="definiens">law of composition</term> on <om:OMOBJ><om:OMV name="E"/></om:OMOBJ>. </CMP> </om:OMOBJ> <om:OMBIND> <om:OMS cd="fns1" name="lambda"/> <om:OMBVAR> <om:OMV name="E"/><om:OMV name="F"/> </om:OMBVAR> <om:OMA><om:OMS cd="pl0" name="and"/> <om:OMA><om:OMS cd="sets" name="set"/> <om:OMV name="E"/> </om:OMA> </pre>

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kohlhase-omdoc_DIA_0020	59	—	—	—	<pre> <om:OMA> <om:OMS cd="functions" name="function" /> <om:OMA id="comp.1"> <om:OMS cd="products" name="Cartesian-product" /> <om:OMV name="E" /> <om:OMV name="E" /> </om:OMA> <om:OMV name="E" /> </om:OMA> </om:OMBJ> </definition> </pre>
kohlhase-omdoc_DIA_0021	59	—	—	—	<pre> <definition xml:id="magma.def" for="#magma" type="implicit"> <CMP> A set with a law of composition is called a <term cd="magmas" name="magma" role="definiens">magma</term>. </CMP> <FMP> <om:OMOBJ> <om:OMBIND><om:OMS cd="p1" name="forall" /> <om:OMBVAR><om:OMV name="M" /></om:OMBVAR> <om:OMA><om:OMS cd="p0" name="iff" /> <om:OMA><om:OMS cd="magmas" name="magma" /> <om:OMV name="M" /> </om:OMA> <om:OMBIND> <om:OMS cd="p1" name="exists" /> <om:OMBVAR> <om:OMV name="E" /><om:OMV name="C" /> </om:OMBVAR> <om:OMA><om:OMS cd="p0" name="and" /> <om:OMA><om:OMS cd="relation1" name="eq" /> <om:OMV name="M" /> <om:OMA><om:OMS cd="products" name="Cartesian-product" /> <om:OMV name="E" /> <om:OMV name="C" /> </om:OMBJ> </FMP> </definition> </pre>
kohlhase-omdoc_DIA_0022	60	—	—	—	<pre> </om:OMA> </om:OMA> <om:OMA><om:OMS cd="magmas" name="law_of_composition" /> <om:OMV name="E" /> <om:OMV name="F" /> </om:OMA> </om:OMA> </om:OMBIND> </om:OMA> </om:OMBIND> </om:OMOBJ> </FMP> </definition> </pre>

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kohlhase-omdoc_DIA_0023	61	—	—	—	<pre> <example xml:id="e11.magma" for="#law_of_composition" type="for" assertion="e11.magma.ass"> <CMP> The mapping <om:OMOBJ><om:OMR href="#e11.magma.1"/></om:OMOBJ> is a law of composition on the set of subsets of a set <om:OMOBJ><om:OMS cd="magmas" name="E"/></om:OMOBJ>. </CMP> <om:OMOBJ> <om:OMA id="e11.magma.2"><om:OMS cd="sets" name="subset"/> <om:OMV name="E"/> </om:OMA> </om:OMOBJ> <om:OMOBJ> <om:OMBIND id="e11.magma.1"> <om:OMS cd="fns1" name="lambda"/> <om:OMBVAR><om:OMV name="X"/><om:OMV name="Y"/></om:OMBVAR> <om:OMA> <om:OMS cd="functions" name="pattern-defined"/> <om:OMA><om:OMS cd="products" name="pair"/> <om:OMV name="X"/> <om:OMV name="Y"/> </om:OMA> <om:OMA><om:OMS cd="sets" name="union"/> <om:OMV name="X"/> <om:OMV name="Y"/> </om:OMA> </om:OMBIND> </om:OMOBJ> </example> <assertion xml:id="e11.magma.ass"> <FMP> <om:OMOBJ> <om:OMBIND> <om:OMS cd="pl1" name="forall"/> <om:OMBVAR><om:OMV name="E"/></om:OMBVAR> <om:OMA> <om:OMS cd="magmas" name="law_of_composition"/> <om:OMR href="#e11.magma.2"/> <om:OMR href="#e11.magma.1"/> </om:OMA> </om:OMBIND> </om:OMOBJ> </FMP> </assertion> </pre>
kohlhase-omdoc_DIA_0024	62	—	—	—	<pre> <CD> <CDName> arith1 </CDName> <CDURL> http://www.openmath.org/cd/arith1.ocd </CDURL> <CDReviewDate> 2003-04-01 </CDReviewDate> <CDStatus> official </CDStatus> <CDDate> 2001-03-12 </CDDate> <CDVersion> 2 </CDVersion> <CDRevision> 0 </CDRevision> <dc:description> This CD defines symbols for common arithmetic functions. </dc:description> <CDDefinition> <Name> lcm </Name> <Description> </pre>
kohlhase-omdoc_DIA_0025	63	—	—	—	<pre> <CMP> lcm(a,b) = a*b/gcd(a,b) </CMP> <FMP>... </FMP> <CMP> for all integers a,b There does not exist a c>0 such that c/a is an Integer and c/b is an Integer and lcm(a,b) > c. </CMP> <FMP>...</FMP> ... </CD> </pre>

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kohlhase-omdoc_DIA_9026	63	—	—	—	<pre> <?xml version="1.0" encoding="utf-8"?> <omdoc xml:id="arith1.omdoc" modules="@cd" xmlns:dc="http://purl.org/dc/elements/1.1/" > <metadata> <dc:title>The OpenMath Content Dictionary arith1.oed in OMDoc Form</dc:title> <dc:creator role="trl">Michael Kohlhase</dc:creator> <dc:creator role="ant">The OpenMath Society</dc:creator> <dc:date action="updated"> 2004-01-17T09:04:03Z </dc:date> <dc:source> Derived from the OpenMath CD http://www.openmath.org/cd/arith1.oed. </dc:source> <dc:type>Text</dc:type> <dc:format>application/omdoc+xml</dc:format> <dc:rights>Copyright (c) 2000 Michael Kohlhase; This OMDoc content dictionary is released under the OpenMath license: </pre>
kohlhase-omdoc_DIA_9027	64	—	—	—	<pre> http://www.openmath.org/cdfiles/license.html </dc:rights> </metadata> <theory xml:id="arith1" cdstatus="official" cdreviewdate="2003-04-01" cdversion="2" cdrevision="0"> <metadata> <dc:title>Common Arithmetic Functions</dc:title> <dc:description>This CD defines symbols for common arithmetic functions.</dc:description> <dc:date action="updated"> 2001-03-12 </dc:date> </metadata> <imports from="#sts"/> <symbol name="lcm"> <metadata> <dc:description>The symbol to represent the n-ary function to return the least common multiple of its arguments. </dc:description> <dc:description xml:lang="de"> Das Symbol für das kleinste gemeinsame Vielfache (als n-äre Funktion). </dc:description> <dc:subject>lcm, least common mean</dc:subject> <dc:subject xml:lang="de">kgV, kleinstes gemeinsames Vielfaches</dc:subject> </metadata> <type system="sts"> <OMOBJ> <OMA><OMS name="mapsto" cd="sts"/> <OMA><OMS name="nassoc" cd="sts"/><OMV name="SemiGroup"/></OMA> <OMV name="SemiGroup"/> </OMA> </OMOBJ> </type> </symbol> <presentation xml:id="pr_lcm" for="#lcm"> <use format="default">lcm</use> <use format="default" xml:lang="de">kgV</use> <use format="cmml" element="lcm"/> </presentation> <definition xml:id="lcm-def" for="#lcm" type="pattern"> <CMP>We define <OMOBJ><OMR href="#lcm-def.O"/></OMOBJ> as <OMOBJ><OMR href="#lcm-def.1"/></OMOBJ></CMP> <CMP xml:lang="de"> Wir definieren <OMOBJ><OMR href="#lcm-def.O"/></OMOBJ> als <OMOBJ><OMR href="#lcm-def.1"/></OMOBJ></CMP> <requation> <OMOBJ> <OMA id="lcm-def.O"> <OMS cd="arith1" name="lcm"/> <OMV name="a"/><OMV name="b"/> </OMA> </OMOBJ> <OMOBJ> <OMA id="lcm-def.1"> <OMS cd="arith1" name="divide"/> <OMA><OMS cd="arith1" name="times"/> <OMV name="a"/> <OMV name="b"/> </OMA> <OMA><OMS cd="arith1" name="gcd"/> <OMV name="a"/> <OMV name="b"/> </OMA> </OMOBJ> </requation> </pre>

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kohlhase-omdoc_DIA_0028	65	—	—	—	<pre> </requation> </definition> <theory> <imports from="#relation1"/> <imports from="#quant1"/> <imports from="#logic1"/> <assertion xml:id="lcm-prop-3" type="lemma"> <CMP>For all integers <OMOBJ><OMV name="a"/></OMOBJ>, <OMOBJ><OMV name="b"/></OMOBJ> there is no <OMOBJ><OMR href="#lcm-prop-3.1"/></OMOBJ> such that <OMOBJ><OMR href="#lcm-prop-3.2"/></OMOBJ> and <OMOBJ><OMR href="#lcm-prop-3.3"/></OMOBJ> and <OMOBJ><OMR href="#lcm-prop-3.4"/></OMOBJ>. </CMP> <CMP xml:lang="de">Für alle ganzen Zahlen <OMOBJ><OMV name="a"/></OMOBJ>, <OMOBJ><OMV name="b"/></OMOBJ> gibt es kein <OMOBJ><OMR href="#lcm-prop-3.1"/></OMOBJ> mit <OMOBJ><OMR href="#lcm-prop-3.2"/></OMOBJ> und <OMOBJ><OMR href="#lcm-prop-3.3"/></OMOBJ> und <OMOBJ><OMR href="#lcm-prop-3.4"/></OMOBJ>. </CMP> <FMP> <OMOBJ><OMBIND><OMS cd="quant1" name="forall"/> <OMBVAR><OMV name="a"/><OMV name="b"/></OMBVAR> <OMA><OMS cd="logic1" name="implies"/> <OMA>...</OMA> <OMA><OMS cd="logic1" name="not"/> <OMBIND><OMS cd="quant1" name="exists"/> <OMBVAR><OMV name="c"/></OMBVAR> <OMA><OMS cd="logic1" name="and"/> <OMA id="lcm-prop-3.1">...</OMA> <OMA id="lcm-prop-3.2">...</OMA> <OMA id="lcm-prop-3.3">...</OMA> <OMA id="lcm-prop-3.4">...</OMA> </OMA> </OMBIND> </OMA> </OMA> </OMBIND> </OMOBJ> </FMP> </assertion> ... </theory> ... </theory> </pre>
kohlhase-omdoc_DIA_0029	67	—	—	—	<pre> graph TD NatOrdList["NatOrdList cons, nil, 0, s, N, <"] OrdList["OrdList cons, nil, Elem, <"] NatOrd["NatOrd 0, s, N, <"] T0Set["T0Set Elem, <"] NatOrd -- imports --> NatOrdList T0Set -- imports --> OrdList NatOrdList -.-> Actualization OrdList NatOrdList -.-> theory-inclusion T0Set NatOrdList -.-> induces T0Set </pre>

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kohlhase-omdoc_DIA_0030	68	—	—	—	<pre> <theory xml:id="TOSet"> <symbol name="set"/> <symbol name="ord"/> <axiom xml:id="toset"><CMP>ord is a total order on set.</CMP></axiom> </theory> <theory xml:id="nat"> <adt> <sortdef name="Nat"> <constructor name="zero"/> <constructor name="succ"> <argument> <type><OMOBJ><OMS name="Nat" cd="nat"/></OMOBJ></type> <selector name="pred"/> </argument> </constructor> </sortdef> </adt> </theory> <theory xml:id="NatOrd"> <imports from="#nat"/> <imports from="#TOSet"/> <symbol name="leq"/> <definition xml:id="leq.def" for="#leq" type="implicit" existence="#leq.ex" uniqueness="#leq.uniq"> <FMP>$\forall x. 0 \leq x \wedge \forall x, y. x \leq y \Rightarrow s(x) \leq s(y)$</FMP> </definition> <assertion xml:id="leq.ex"><CMP>$\leq$ exists.</CMP></assertion> <assertion xml:id="leq.unique"><CMP>$\leq$ is unique</CMP></assertion> <assertion xml:id="leq.TO"><CMP>$\leq$ is a total order on Nat.</CMP></assertion> </theory> </pre>
kohlhase-omdoc_DIA_0031	69	—	—	—	<pre> <theory-inclusion xml:id="elem-nat-incl" to="#NatOrd" from="#TOSet"> <morphism xml:id="elem-nat" type="pattern"> <requation> <OMOBJ><OMS cd="TOSet" name="set"/></OMOBJ> <OMOBJ><OMS cd="NatOrd" name="Nat"/></OMOBJ> </requation> <requation> <OMOBJ><OMS cd="TOSet" name="ord"/></OMOBJ> <OMOBJ><OMS cd="NatOrd" name="leq"/></OMOBJ> </requation> </morphism> <obligation induced-by="#toset" assertion="#leq.TO"/> </theory-inclusion> </pre>
kohlhase-omdoc_DIA_0032	69	—	—	—	<pre> <theory xml:id="OrdList"> <imports from="#TOSet"/> <adt xml:id="list-adt"> <sortdef name="lists"> <constructor name="cons"> <argument> <type><OMOBJ><OMS name="set" cd="TOSet"/></OMOBJ></type> <selector name="head"/> </argument> <argument> <type><OMOBJ><OMS name="lists" cd="OrdList"/></OMOBJ></type> <selector name="rest"/> </argument> </constructor> <constructor name="nil"/> </sortdef> </adt> <symbol name="ordered"/> <definition xml:id="ordered-def" for="#ordered" type="informal"> <CMP>A list l is called ordered, iff $head(l) \leq z$ for all elements $z \in rest(l)$ and $rest(l)$ is ordered.</CMP> </definition> </theory> </pre>
kohlhase-omdoc_DIA_0033	70	—	—	—	<pre> <theory xml:id="NatOrdList"> <imports xml:id="natordlist.im-natord" from="#NatOrd"/> <imports xml:id="natordlist.im-elt" from="#OrdList" type="local"> <morphism base="#elem-nat"/> </imports> <inclusion via="elem-nat-incl"/> </theory> </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0034	70	—	—	—	<pre> <axiom-inclusion xml:id="toset-natordlist-incl" from="#TOSet" to="#NatOrdList"> <morphism base="#elem-nat"/> <path-just local="#elem-nat-incl" globals="#natordlist.im-natord"/> </axiom-inclusion> <theory-inclusion from="#OrdList" to="#NatOrdList"> <morphism base="#elem-nat"/> <decomposition links="#toset-natordlist-incl #elem-nat-incl"/> </theory-inclusion> </pre>
kohlhase-omdoc_DIA_0035	71	—	—	—	
kohlhase-omdoc_DIA_0036	72	—	—	—	<pre> <theory xml:id="semigroup"> <symbol name="base-set"/> <presentation for="#base-set"><use format="default">M</use></presentation> <symbol name="op"/> <presentation for="#op"><use format="default">o</use></presentation> <axiom xml:id="closed.ax"><FMP>∀x,y ∈ M.x o y ∈ M</FMP></axiom> <axiom xml:id="assoc.ax"> <FMP>∀x,y,z ∈ M.(x o y) o z = x o (y o z)</FMP> </axiom> </theory> <theory-inclusion xml:id="sg-conv-sg" from="#semigroup" to="#semigroup"> <morphism xml:id="sg-conv-sg.morphism"> <equation>X o Y ~ Y o X</equation> </morphism> <obligation assertion="conv.closed" induced-by="#closed.ax"/> <obligation assertion="#assoc.ax" induced-by="#assoc.ax"/> </theory-inclusion> </pre>
kohlhase-omdoc_DIA_0037	72	—	—	—	<pre> <theory xml:id="monoid"> <imports xml:id="sg2mon" from="#semigroup"/> <axiom xml:id="unit.ax"><FMP>∃x ∈ M.∀y ∈ M.y o x = y</FMP></axiom> <assertion xml:id="unit.unique"><FMP>∃!x ∈ M.∀y ∈ M.y o x = y</FMP></assertion> <symbol name="unit"/> <presentation for="#unit"><use format="default">e</use></presentation> <definition xml:id="unit.def" for="#unit" type="simple" existence="#unit.unique"> τx ∈ M.∀y ∈ M.y o x = y </definition> <assertion xml:id="left.unit"><FMP>∀x ∈ M.e o x = x</FMP></assertion> <symbol name="setstar"/> <presentation for="#setstar" fixity="postfix"> <use format="default">*</use> </presentation> <definition xml:id="ss.def" for="#setstar" type="implicit"> ∀S ⊆ M.S* = S \ {e} </definition> </theory> </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0038	73	—	—	—	<pre> <axiom-inclusion xml:id="mon-conv-mon.local" from="#monoid" to="#monoid"> <morphism base="#sg-conv-sg.morphism"/> <obligation assertion="#left.unit" induced-by="#unit.ax"/> </axiom-inclusion> <axiom-inclusion xml:id="sg-conv-mon" from="#semigroup" to="#monoid"> <morphism base="#sg-conv-sg.morphism"/> <path-just local="#sg-conv-sg" globals="#sg2mon"/> </axiom-inclusion> <theory-inclusion xml:id="mon-conv-mon.global" from="#monoid" to="#monoid"> <morphism base="#sg-conv-sg.morphism"/> <decomposition links="#sg-conv-sg #sg-conv-mon"/> </theory-inclusion> </pre>
kohlhase-omdoc_DIA_0039	73	—	—	—	<pre> <theory xml:id="group"> <imports xml:id="mon2grp" from="#monoid"/> <axiom xml:id="inv.ax"><FMP>$\forall x \in M. \exists y \in M. x \circ y = e$</FMP></axiom> <symbol name="inv"/> <presentation for="#inv" role="applied"> <use format="default" lbrack=" " rbrack=" " fixity="postfix">$^{-1}$</use> </presentation> <definition xml:id="inv.def" for="#inv" type="pattern"> <requation>$x^{-1} \rightsquigarrow \tau y. x \circ y = e$</value></requation> </definition> <assertion xml:id="conv.inv"><FMP>$\forall x \in M. \exists y \in M. y \circ x = e$</FMP></assertion> </theory> </pre>
kohlhase-omdoc_DIA_0040	73	—	—	—	<pre> <axiom-inclusion xml:id="grp-conv-grp.local" from="#group" to="#group"> <morphism base="#sg-conv-sg.morphism"/> <obligation assertion="conv.inv" induced-by="#inv.ax"/> </axiom-inclusion> <axiom-inclusion xml:id="sg-conv-grp" from="#semigroup" to="#group"> <morphism base="#sg-conv-sg.morphism"/> <path-just local="#sg-conv-sg" globals="#mon2grp #sg2mon"/> </axiom-inclusion> <axiom-inclusion xml:id="mon-conv-grp" from="#monoid" to="#group"> <morphism base="#sg-conv-sg.morphism"/> <path-just local="#mon-conv-mon.local" globals="#mon2grp"/> </axiom-inclusion> <theory-inclusion xml:id="grp-conv-grp" from="#group" to="#group"> <morphism base="#sg-conv-sg.morphism"/> <decomposition links="#sg-conv-grp #mon-conv-grp #grp-conv-grp.local"/> </theory-inclusion> </pre>
kohlhase-omdoc_DIA_0041	74	—	—	—	<pre> <theory xml:id="ring"> <symbol name="R"/> <presentation for="#R"><use format="default">R</use></presentation> <symbol name="zero"/> <presentation for="#plus" role="applied"> <use format="default">+</use> </presentation> <symbol name="one"/> <presentation for="#zero"><use format="default">0</use></presentation> <symbol name="times"/> <presentation for="#negative" role="applied"> <use format="default">-</use> </presentation> <symbol name="times"/> <presentation for="#times" role="applied"> <use format="default">*</use> </presentation> <symbol name="one"/> <presentation for="#one"><use format="default">1</use></presentation> <imports xml:id="add.import" from="#group"> <morphism>$M \mapsto R, x \circ y \mapsto x * y, e \mapsto 1, \cdot^{-1} \mapsto -$</morphism> </imports> <imports xml:id="mult.import" from="#monoid"> <morphism hiding="setstar">$M \mapsto M^*, x \circ y \mapsto x * y, e \mapsto 1$</morphism> </imports> <axiom xml:id="dist.ax"><FMP>$x * (y + z) = (x * y) + (x * z)$</FMP></axiom> <assertion xml:id="dist.conv"><FMP>$(z + y) * x = (z * x) + (y * x)$</FMP></assertion> </theory> </pre>

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kohlhase-omdoc_DIA_0042	74	—	—	—	<pre> <axiom-inclusion xml:id="sg-conv-rg.add" from="#semigroup" to="#ring"> <morphism base="#sg-conv-sg.morphism #add.import"/> <path-just local="#sg-conv-sg" globals="#sg2mon #mon2grp #add.import"/> </axiom-inclusion> <axiom-inclusion xml:id="mon-conv-rg.add" from="#monoid" to="#group"> <morphism base="#sg-conv-sg.morphism #add.import"/> <path-just local="#mon-conv-mon.local" globals="#mon2grp #add.import"/> </axiom-inclusion> <axiom-inclusion xml:id="grp-conv-rg.add" from="#group" to="#group"> <morphism base="#sg-conv-sg.morphism #add.import"/> <path-just local="#grp-conv-grp.local" globals="#add.import"/> </axiom-inclusion> </pre>
kohlhase-omdoc_DIA_0043	75	—	—	—	<pre> <axiom-inclusion xml:id="rg-conv-rg.local" from="#ring" to="#ring"> <morphism xml:id="rg-conv-rg.morphism">$x + y \mapsto y + x, x * y \mapsto y * x$</morphism> <obligation assertion="#dist.conv" induced-by="#dist.ax"/> </axiom-inclusion> <theory-inclusion xml:id="rg-conv-rg" from="#ring" to="#ring"> <morphism base="#rg-conv-rg.morphism"/> <decomposition links="#rg-conv-rg.local #sg-conv-rg.add #mon-conv-rg.add #grp-conv-rg.add #sg-conv-rg.mult #mon-conv-rg.mult #grp-conv-rg.mult"/> </theory-inclusion> </pre>
kohlhase-omdoc_DIA_0044	76	—	—	—	Inheritance: Taxonomy metaphor Classes vs instances - Every <i>object</i> is an <i>instance</i> of a <i>class</i>. - The characteristics of an object are defined by its class. - An object <i>inherits</i> characteristics from all of its <i>superclasses</i>. Classes vs instances Example: <i>Danny Sleator</i> is an <i>instance</i> of the <i>Professor</i> class. - He is therefore also an instance of the <i>Human</i>, <i>Mammal</i>, and <i>Animal</i> classes. - Sometimes we say that Danny Sleator <i>"is a"</i> Professor (or Human or Mammal...) - Danny also <i>"has a"</i> wife and son, who are also instances of the Human class. In Java: <pre> public class Animal { ... } <i>Implicitly extends class Object</i> public class Mammal extends Animal { ... } public class Human extends Mammal { ... } public class Professor extends Human { ... } public class MyClass { public static main () { Professor danny = new Professor(); } } </pre>
kohlhase-omdoc_DIA_0045	78	—	—	—	<pre> <theory xml:id="animals-tax"> <imports xml:id="tax_imports_taxonomy" from="#taxonomies"/> <imports xml:id="tax_imports_kr" from="#kr"/> <symbol name="human"> <type system="stlc"><OMOBJ><OMS cd="kr" name="concept"/></OMOBJ></type> </symbol> <symbol name="mammal"> <type system="stlc"><OMOBJ><OMS cd="kr" name="concept"/></OMOBJ></type> </symbol> ... <axiom xml:id="mammal-ext-human"> <CMP>Humans are Animals.</CMP> <FMP> <OMOBJ> <OMA><OMS cd="kr" name="extends"/> <OMS cd="animal-taxonomy" name="mammal"/> <OMS cd="animal-taxonomy" name="human"/> </OMA> </OMOBJ> </FMP> </axiom> ... </theory> <private xml:id="tax-image" for="#animals-tax" reformulates="#animals-tax"> <data format="image/jpeg" href="animals-taxonomy.jpg"/> <data format="application/postscript" href="animals-taxonomy.ps"/> </private> </pre>

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kohlhase-omdoc_DIA_0046	79	—	—	—	<pre> <theory xml:id="cvi"> <symbol name="object" xml:id="cvi.object"/> <symbol name="instance" xml:id="cvi.instance"/> <symbol name="class" xml:id="cvi.class"/> <symbol name="inherits" xml:id="cvi.inherits"/> <symbol name="superclass" xml:id="cvi.superclass"/> <axiom xml:id="ax1" for="#cvi.object #cvi.instance #cvi.class"> <CMP>Every <phrase style="font-style:italic;color:blue">object</phrase> is an <phrase style="font-style:italic; color:red">instance</phrase> of a <phrase style="font-style:italic; color:blue">class</phrase>. </CMP> </axiom> <axiom xml:id="ax2" for="#cvi.class"> <CMP>The characteristics of an object are defined by its class.</CMP> </axiom> <axiom xml:id="ax3" for="#cvi.inherits #cvi.superclass"> <CMP> An object <phrase style="font-style:italic;color:blue">inherits</phrase> characteristics from all of its <phrase style="font-style:italic; color:red">superclasses</phrase>.</CMP> </axiom> </theory> </pre>
kohlhase-omdoc_DIA_0047	80	—	—	—	<pre> <theory xml:id="cvi-examples"> <imports from="#animals-tax"/><imports from="#cvi"/> <symbol name="danny" xml:id="cvi-examples.danny"> <metadata><dc:description>Danny Sleator</dc:description></metadata> </symbol> <axiom xml:id="danny-professor" for="#cvi.class #cvi.instance #cvi-examples.danny"> <CMP><phrase style="font-style:italic;color:blue">Danny Sleator</phrase> is an <phrase style="font-style:italic; color:red">instance</phrase> of the <phrase style="font-style:italic; color:blue">Professor</phrase> class. </CMP> </axiom> <assertion xml:id="dannys-classes" type="theorem"> <CMP>He is therefore also an instance of the <phrase style="font-style:italic; color:blue">Human</phrase>, <phrase style="font-style:italic; color:blue">Mammal</phrase>, <phrase style="font-style:italic; color:blue">Animal</phrase> classes. </CMP> </assertion> <symbol name="is_a" scope="global"> <metadata><dc:subject>'is a' relation</dc:subject></metadata> </symbol> <definition xml:id="is_a-def" for="#is_a" type="informal"> <CMP>Sometimes we say that Danny Sleator &#x201C;<phrase style="font-style:italic;color:red">is a</phrase>&#x201D; Professor (or Human or Mammal&#x2026;) </CMP> </definition> </theory> </pre>

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kohlhase-omdoc_DIA_0048	81	—	—	—	... <example xml:id="danny-mammal" type="for" assertion="dannys-mammal-thm" for="#cvi.instance #cvi.class #animal-taxonomy.mammal"> <CMP>Danny Sleator is an instance of the <phrase style="font-style:italic; color:blue">Mammal</phrase> class. </CMP> <OMOBJ><OMS cd="cvi-examples" name="danny"/></OMOBJ> </example> <assertion xml:id="dannys-mammal-thm" type="theorem" proofs="danny-mammal-pf"> <CMP>Danny Sleator is an instance of the Human class.</CMP> </assertion> <proof xml:id="danny-human-pf" for="#dannys-mammal-thm"> <derive xml:id="d1"> <CMP>Danny Sleator is an instance of the human class.</CMP> <method> <premise xref="#danny-professor"/> <premise xref="#cvi.ax3"/> <premise xref="#animal-tax.human-extends-professor"/> </method> </derive> <derive xml:id="concl"> <CMP>Therefore he is an instance of the human class.</CMP> <method> <premise xref="#d1"/> <premise xref="#cvi.ax3"/> <premise xref="#animal-tax.mammal-extends-human"/> </method> </derive> </proof> ...
kohlhase-omdoc_DIA_0049	82	—	—	—	<code xml:id="cvic-code1"> <data format="Java"><![CDATA[public class Animal {...}]]></data> <effect><CMP>class declaration</CMP></effect> </code> <code xml:id="cvic-code2" requires="cvic-code1"> <data format="Java"><![CDATA[public class Mammal extends Animal {...}]]></data> <effect><CMP>class extension</CMP></effect> </code> ...

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kohlhase-omdoc_DIA_0050	83	—	—	—	<pre> ... <omgroup xml:id="slide-847" type="slide"> <metadata> <dc:title class="15-211-title">Inheritance: Taxonomy metaphor</dc:title> </metadata> <omtext xml:id="the-tax"> <CMP> <omlet data="#tax-image" style="width:540;height:366" action="display" show="embed" /> </CMP> </omtext> </omgroup> <omgroup xml:id="slide-848" type="slide"> <metadata><dc:title class="15-211-title">Classes vs. instances</dc:title></metadata> <omgroup type="itemize" style="list-style-type:url(square.gif)"> <ref style="position:30% 10%" xml:id="obj" xref="slide1.content.omdoc#ax1" /> <ref style="position:55% 10%" xml:id="class" xref="slide1.content.omdoc#ax2" /> <ref style="position:80% 10%" xml:id="inh" xref="slide1.content.omdoc#ax3" /> </omgroup> </omgroup> <omgroup xml:id="slide-849" type="slide"> <metadata><dc:title class="15-211-title">Classes vs. instances</dc:title></metadata> <omgroup type="itemize" style="list-style-type:url(square.gif)"> <omtext style="position:30% 10%" xml:id="ex"><CMP>Example:</CMP></omtext> <omgroup type="itemize" style="list-style-type:url(triangle.gif)"> <ref style="position:400% 15%" xml:id="danny" xref="slide1.content.omdoc#danny-professor" /> <ref style="position:55% 15%" xml:id="inst" xref="slide1.content.omdoc#dannys-classes" /> <ref style="position:70% 15%" xml:id="is_a" xref="slide1.content.omdoc#is_a-def" /> </omgroup> <omtext style="position:83% 10%" xml:id="has_a"> <CMP> Danny also &#x201C;<phrase style="font-style:italic;color:red">has a</phrase>&#x201D; wife and son, who are also instances of the Human class </CMP> </omtext> </omgroup> </omgroup> <omgroup xml:id="slide-850" type="slide"> <metadata><dc:title class="15-211-title">In Java</dc:title></metadata> <omgroup type="itemize"> <omtext xml:id="slide-850.t1" style="position:80% 10%;color:red"> <CMP>Implicitly extends class object</CMP> </omtext> <omtext xml:id="slide-850.t2"> <CMP><omlet data="#cvic-code1" action="display" show="embed" /></CMP> </omtext> <omtext xml:id="slide-850.t3"> <CMP><omlet data="#cvic-code2" action="display" show="embed" /></CMP> </omtext> </omgroup> </omgroup> ... </pre>
kohlhase-omdoc_DIA_0051	84	—	—	—	

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kohlhase-omdoc_DIA_0052	87	—	—	—	HTTP/1.1 200 OK Content-Type: application/soap+xml Content-Length: 990 <pre> <?xml version='1.0'?> <env:Envelope xmlns:env="http://www.w3.org/2003/05/soap-envelope"> <env:Body> <ws:prover env:encodingStyle="http://www.w3.org/2003/05/soap-encoding" xmlns:ws="http://www.mathweb.org/ws-fictional"> <ws:name>SPASS</ws:name> <ws:version>2.1</ws:version> <ws:URL>http://spass.mpi-sb.mpg.de/webspass/soap</ws:URL> <ws:uptime>P3D5H6M45S</ws:uptime> <ws:sysinfo> <ws:ostype>SunOS 5.6</ws:ostype> <ws:mips>3825</ws:mips> </ws:sysinfo> </ws:prover> <ws:prover env:encodingStyle="http://www.w3.org/2003/05/soap-encoding" xmlns:ws="http://www.mathweb.org/ws-fictional"> <ws:name>SPASS</ws:name> <ws:version>2.0</ws:version> <ws:URL>http://asuka.mt.cs.cmu.edu/atp/spass/soap</ws:URL> <ws:uptime>P5M2D15H56M5S</ws:uptime> <ws:sysinfo> <ws:ostype>linux-2.4.20</ws:ostype> <ws:mips>1468</ws:mips> </ws:sysinfo> <ws:prover> </env:Body> </end:Envelope> </pre>
kohlhase-omdoc_DIA_0053	87	—	—	—	POST http://spass.mpi-sb.mpg.de/webspass/soap HTTP/1.1 Host: http://spass.mpi-sb.mpg.de/webspass/soap Content-Type: application/soap+xml; Content-Length: 1123 <pre> <?xml version='1.0'?> <env:Envelope xmlns:env="http://www.w3.org/2003/05/soap-envelope"> <env:Body> <ws:prove env:encodingStyle="http://www.w3.org/2003/05/soap-encoding" xmlns:ws="http://www.mathweb.org/ws-fictional"> <omdoc:assertion xml:id="peter-hates-somebody" type="conjecture" xmlns:omdoc="http://www.mathweb.org/omdoc" theory="http://mbase.mathweb.org:8080/RPC2#lovelife"> <omdoc:CMP>Peter hates somebody</omdoc:CMP> </ws:prove> </env:Body> </end:Envelope> </pre>
kohlhase-omdoc_DIA_0054	88	—	—	—	<pre> <omdoc:FMP> <om:OMOBJ xmlns:om="http://www.openmath.org/OpenMath"> <om:OMBIND> <om:OMS cd="quant1" name="exists"/> <om:OMBVAR><om:OMV name="X"/></om:OMBVAR> <om:OMA> <om:OMS cd="lovelife" name="hate"/> <om:OMS cd="lovelife" name="peter"/> <om:OMV name="X"/> </om:OMA> </om:OMBIND> </om:OMOBJ> </omdoc:FMP> </omdoc:assertion> <ws:replyWith><ws:state>proof</ws:state></ws:replyWith> <ws:timeout>20</ws:timeout> </ws:prove> </env:Body> </env:Envelope> </pre>
kohlhase-omdoc_DIA_0055	89	—	—	—	GET /mbase.mathweb.org:8080/soap/getTheory?name=lovelife HTTP/1.1 Host: mbase.mathweb.org:8080 Accept: application/soap+xml

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kohlhase-omdoc_DIA_0056	89	—	—	—	HTTP/1.1 200 OK Content-Type: application/soap+xml Content-Length: 602 <pre> <?xml version='1.0'?> <env:Envelope xmlns:env="http://www.w3.org/2003/05/soap-envelope"> <env:Body> <theory xml:id="lovelife" xmlns="http://www.mathweb.org/omdoc"> <symbol name="peter"/><symbol name="mary"/> <symbol name="love"/><symbol name="hate"/> <axiom xml:id="opposite"> <CMP>Peter hates everybody Mary loves</CMP> <FMP>$\forall x. loves(mary, x) \Rightarrow hates(peter, x)$</FMP> </axiom> <axiom xml:id="mary-loves-peter"> <CMP>Mary loves Peter</CMP> <FMP>$loves(mary, peter)$</FMP> </axiom> </theory> </env:Body> </env:Envelope> </pre>
kohlhase-omdoc_DIA_0057	89	—	—	—	HTTP/1.1 200 OK Content-Type: application/soap+xml Content-Length: 588 <pre> <?xml version='1.0'?> <env:Envelope xmlns:env="http://www.w3.org/2003/05/soap-envelope"> <env:Body> <proof xml:id="p347" for="#peter-hates-somebody" xmlns="http://www.mathweb.org/omdoc"> <derive xml:id="d1"> <FMP>$hates(peter, peter)$</FMP> <method xref="nd.omdoc#ND.chain"> <premise xref="#lovelife.mary-loves-peter"/> <premise xref="#lovelife.opposite"/> </method> </derive> <derive xml:id="concl"> <method xref="nd.omdoc#ND.Ex1"><premise xref="#d1"/></method> </derive> </proof> </env:Body> </env:Envelope> </pre>
kohlhase-omdoc_DIA_0058	94	—	—	—	<pre> <?xml version="1.0" encoding="utf-8"?> <omdoc xml:id="test.omdoc" version="1.2" xmlns="http://www.mathweb.org/omdoc" xmlns:cc="http://creativecommons.org/ns" xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:om="http://www.openmath.org/OpenMath" xmlns:m="http://www.w3.org/1998/Math/MathML"> ... </omdoc> </pre>
kohlhase-omdoc_DIA_0059	95	—	—	—	<pre> <?xml version="1.0" encoding="utf-8"?> <?xml-stylesheet type="text/css" href="http://example.org/style.css"?> <omdoc xml:id="stylish"> ... <omtext xml:id="t1" style="width:80%;align:center;border:2px #006699 solid"> <CMP>Here comes something <phrase style="font-weight:bold;color:green" class="emphasize">stylish</phrase>! </CMP> </omtext> ... </omdoc> </pre>
kohlhase-omdoc_DIA_0060	101	—	—	—	<pre> <ignore type="todo" comment="this does not make sense yet, rework"> <assertion xml:id="heureka">...</assertion> </ignore> </pre>
kohlhase-omdoc_DIA_0061	101	—	—	—	<pre> <ignore type="todo" comment="rework"> <ignore type="todo-comment"> <CMP>This does not make sense yet, in particular, the equation <OMOBJ>...</OMOBJ> cannot be true, think of <OMOBJ>...</OMOBJ> </CMP> </ignore> <assertion xml:id="heureka">...</assertion> </ignore> </pre>

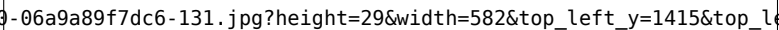
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kohlhase-omdoc_DIA_0062	101	—	—	—	<pre> <ignore type="CMP–lang–error" comment="multilingual CMPs are not translations of each other"> <assertion xml:id="ass1"> <CMP>The proof is trivial</CMP> <CMP xml:lang="de">Der Beweis ist extrem schwer</CMP> </assertion> </ignore> </pre>
kohlhase-omdoc_DIA_0063	104	—	—	—	<pre> <omgroup xml:id="text" type="sequence"> <omtext xml:id="t1">T_1</omtext> <omgroup xml:id="enum" type="enumeration"> <omtext xml:id="t2">T_2</omtext> <omtext xml:id="t3">T_3</omtext> </omgroup> <omtext xml:id="t4">T_4</omtext> </omgroup> </pre> ↔
kohlhase-omdoc_DIA_0064	104	—	—	—	<pre> <omgroup xml:id="text" type="sequence"> <ref xref="#t1"/> <ref xref="#enum"/> <ref xref="#t4"/> </omgroup> <ignore type="targets" comment="already referenced"> <omtext xml:id="t1">T_1</omtext> <omtext xml:id="t2">T_2</omtext> <omtext xml:id="t3">T_3</omtext> <omtext xml:id="t4">T_4</omtext> <omgroup xml:id="enum" type="enumeration"> <ref xref="#t2"/> <ref xref="#t3"/> </omgroup> </ignore> </pre>
kohlhase-omdoc_DIA_0065	109	—	—	—	<pre> <metadata> <dc:title>The Joy of Jordan C^* Triples</dc:title> <dc:creator role="aut">A</dc:creator> <dc:contributor role="edt">R</dc:contributor> <dc:contributor role="trc">S</dc:contributor> </metadata> </pre>
kohlhase-omdoc_DIA_0066	110	—	—	—	<pre> <omdoc xml:id="NNat" version="1.2" xmlns:dc="http://purl.org/dc/elements/1.1/"> <metadata><dc:title>Natural Numbers</dc:title></metadata> ... <theory xml:id="NNat.thy"> <metadata> <dc:title>Natural Numbers</dc:title> <dc:creator role="aut">R</dc:creator> <dc:contributor role="ant">A</dc:contributor> <dc:source>B</dc:source> </metadata> ... </theory> ... </omdoc> </pre>
kohlhase-omdoc_DIA_0067	112	—	—	—	<pre> <metadata> <dc:rights>Copyright (c) 2004 Michael Kohlhase</dc:rights> <license jurisdiction="de" xmlns="http://creativecommons.org/ns"> <permissions reproduction="permitted" distribution="permitted" derivative_works="permitted"/> <prohibitions commercial_use="permitted"/> <requirements notice="required" copyleft="required" attribution="required"/> </license> </metadata> </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0068	113	—	—	—	<pre> <omdoc xml:id="o1"> <metadata> <dc:creator>MiKo</dc:creator> </metadata> <theory xml:id="th1"> <symbol name="s1"/> <definition for="#s1" xml:id="d1"/> </theory> <theory xml:id="th2"> <metadata> <dc:creator>Paul</dc:creator> </metadata> <symbol name="s2"/> <definition for="#s2" xml:id="d1"> <metadata> <dc:creator>MiKo</dc:creator> </metadata> </definition> </theory> </omdoc> </pre> ↔ <pre> <omdoc xml:id="o1"> <metadata> <dc:creator>MiKo</dc:creator> </metadata> <theory xml:id="th1"> <metadata> <dc:creator>MiKo</dc:creator> </metadata> <symbol name="s1"/> <definition for="#s1" xml:id="d1"> <metadata> <dc:creator>MiKo</dc:creator> </metadata> </definition> </theory> <theory xml:id="th2"> <metadata> <dc:creator>Paul</dc:creator> </metadata> <symbol name="s2"/> <definition for="#s2" xml:id="d1"> <metadata> <dc:creator>MiKo</dc:creator> </metadata> </definition> </theory> </omdoc> </pre>
kohlhase-omdoc_DIA_0069	116	—	—	—	<pre> <OMOBJ xmlns="http://www.openmath.org/OpenMath"> <OMA cdbase="http://www.openmath.org/cd"> <OMS cd="transc1" name="sin"/> <OMV name="x"/> </OMA> </OMOBJ> </pre>
kohlhase-omdoc_DIA_0070	116	—	—	—	<pre> <OMOBJ cdbase="http://www.openmath.org/cd"> <OMBIND> <OMS cd="quant1" name="forall"/> <OMBVAR><OMV name="x"/></OMBVAR> <OMA> <OMS cd="arith1" name="leq"/> <OMA><OMS cd="transc1" name="sin"/><OMV name="x"/></OMA> <OMS cd="nums1" name="pi"/> </OMA> </OMBIND> </OMOBJ> </pre>
kohlhase-omdoc_DIA_0071	117	—	—	—	<pre> <OMATTR> <OMATP> <OMS cd="alt-rep" name="ascii"/> <OMSTR>(x+1)</OMSTR> <OMS cd="alt-rep" name="svg"/> <OMFOREIGN encoding="application/svg+xml"> <svg xmlns='http://www.w3.org/2000/svg'>...</svg> </OMFOREIGN> <OMS cd="pres" name="color"/> <OMS cd="pres" name="red"/> </OMATP> <OMA> <OMS cd="arith1" name="plus"/> <OMV name="x"/> <OMS cd="nums1" name="pi"/> </OMA> </OMATTR> </pre>
kohlhase-omdoc_DIA_0072	118	—	—	—	<pre> <OME> <OMS cd="type-error" name="arity-violation"/> <OMA> <OMS cd="transc1" name="sin"/> <OME> <OMS cd="error" name="unexpected_symbol"/> <OMS cd="nums1" name="po"/> </OME> <OMV name="x"/> </OMA> </OME> </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0073	119	—	—	—	Tree $2^d - 1$ nodes DAG d nodes

Identifier	Page	Conf.	LaTeX source	Rendered	Image
					OPENMATH <pre> <OMOBJ> <OMBIND> <OMS cd="quant1" name="forall" /> <OMBVAR> <OMATTR> <OMATP> <OMS cd="sts" name="type" /> <OMS cd="setname1" name="R" /> </OMATP> <OMV name="a" /> </OMATTR> <OMATTR> <OMATP> <OMS cd="sts" name="type" /> <OMS cd="setname1" name="R" /> </OMATP> <OMV name="b" /> </OMATTR> </OMBVAR> <OMA> <OMS cd="relation" name="eq" /> <OMA> <OMS cd="arith1" name="plus" /> <OMV name="a" /> <OMV name="b" /> </OMA> <OMA> <OMS cd="arith1" name="plus" /> <OMV name="b" /> <OMV name="a" /> </OMA> </OMA> </OMBIND> </OMOBJ> </pre> MATHML <pre> <m:math> <m:apply> <m:forall/> <m:bvar> <m:ci type="real">a</m:ci> </m:bvar> <m:bvar> <m:ci type="real">b</m:ci> </m:bvar> <m:apply> <m:eq/> <m:apply> <m:plus/> <m:ci type="real">a</m:ci> <m:ci type="real">b</m:ci> </m:apply> <m:apply> <m:plus/> <m:ci type="real">b</m:ci> <m:ci type="real">a</m:ci> </m:apply> </m:apply> </m:math> </pre>
kohlhase-omdoc_DIA_0076	124	—	—	—	
					<pre> <OMOBJ> <OMBIND> <OMS cd="quant1" name="forall" /> <OMBVAR> <OMATTR> <OMATP> <OMS cd="sts" name="type" /> <OMA><OMS cd="sts" name="mapsto" /> <OMA><OMS cd="sts" name="mapsto" /> <OMS cd="setname1" name="R" /> <OMS cd="setname1" name="R" /> </OMA> <OMA><OMS cd="sts" name="mapsto" /> <OMS cd="setname1" name="R" /> <OMS cd="setname1" name="R" /> </OMA> </OMATP> </OMATTR> <OMV name="F" /> </OMBVAR> ... </OMBIND> </OMOBJ> </pre>
kohlhase-omdoc_DIA_0077	125	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0078	125	—	—	—	<pre> <m:math> <m:apply> <m:forall/> <m:bvar> <m:semantics> <m:ci>F</m:ci> <m:annotation-xml definitionURL="http://www.openmath.org/cd/sts#type"> <m:apply> <m:csymbol definitionURL="http://www.openmath.org/cd/sts#mapsto"/> <m:apply> <m:csymbol definitionURL="http://www.openmath.org/cd/sts#mapsto"/> <m:csymbol definitionURL="http://www.openmath.org/cd/setname1#real"/> <m:csymbol definitionURL="http://www.openmath.org/cd/setname1#real"/> </m:apply> <m:apply> <m:csymbol definitionURL="http://www.openmath.org/cd/sts#mapsto"/> <m:csymbol definitionURL="http://www.openmath.org/cd/setname1#real"/> <m:csymbol definitionURL="http://www.openmath.org/cd/setname1#real"/> </m:apply> </m:apply> </m:annotation-xml> </m:semantics> </m:bvar> ... </m:apply> </m:math> </pre>
kohlhase-omdoc_DIA_0079	127	—	—	—	<pre> <m:lambda> <m:bvar><m:ci xml:id="the-boundvar">complex presentation</m:ci></m:bvar> <m:apply> <m:plus/> <m:ci definitionURL="#the-boundvar">complex presentation</m:ci> <m:ci definitionURL="#the-boundvar">complex presentation</m:ci> </m:apply> </m:lambda> </pre>
kohlhase-omdoc_DIA_0080	127	—	—	—	<pre> <legacy format="TeX" formalism="http://en.wikipedia.org/wiki/Lambda_calculus"> \lambda{x}{x} </legacy> </pre>
kohlhase-omdoc_DIA_0081	129	—	—	—	<pre> <CMP> Let <OMOBJ id="set"><OMV name="V"/></OMOBJ> be a set. A <term role="definiens">unary operation</term> on <OMOBJ><OMR href="#set"/></OMOBJ> is a function <OMOBJ id="fun"><OMV name="F"/></OMOBJ> with <OMOBJ id="im"> <OMA> <OMS cd="relations1" name="eq"/> <OMA><OMS cd="fns1" name="domain"/><OMV name="F"/></OMA> <OMV name="V"/> </OMA> </OMOBJ>. </CMP> </pre>
kohlhase-omdoc_DIA_0082	130	—	—	—	<pre> </OMOBJ> and <OMOBJ id="ran"> <OMA> <OMS cd="relations1" name="eq"/> <OMA><OMS cd="fns1" name="range"/><OMV name="F"/></OMA> <OMV name="V"/> </OMA> </OMOBJ>. </CMP> <CMP xml:lang="de"> Sei <OMOBJ><OMR href="#set"/></OMOBJ> eine Menge. Eine <term role="definiens">unäre Operation</term> ist eine Funktion <OMOBJ><OMR href="#fun"/></OMOBJ>, so dass <OMOBJ><OMR href="#im"/></OMOBJ> und <OMOBJ><OMR href="#ran"/></OMOBJ>. </CMP> <CMP xml:lang="fr"> Soit <OMOBJ><OMR href="#set"/></OMOBJ> un ensemble. Une <term role="definiens">opération unaire</term> sûr <OMOBJ><OMR href="#set"/></OMOBJ> est une fonction <OMOBJ><OMR href="#fun"/></OMOBJ> avec <OMOBJ><OMR href="#im"/></OMOBJ> et <OMOBJ><OMR href="#ran"/></OMOBJ>. </CMP> </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0083	131	—	—	—	<pre> <omtext xml:id="binop-def" type="definition"> ... the content of Listing 14.3 here ... <FMP logic="fol">\forall V, F. binop(F, V) \Leftrightarrow \mathbf{Im}(F) = V \wedge \mathbf{Dom}(F) = V</FMP> <FMP logic="hol">binop = \lambda V, F. \mathbf{Im}(F) = V \wedge \mathbf{Dom}(F) = V</FMP> </omtext> </pre>
kohlhase-omdoc_DIA_0084	131	—	—	—	<pre> <FMP><conclusion>A</conclusion></FMP> </pre>
kohlhase-omdoc_DIA_0085	131	—	<pre> <pre> \` \` \` <phrase index="moon_cheese">the moon is made of green cheese</phrase>. </CMP> <FMP> <assumption xml:id=" A" > a E U</assumption> <assumption xml:id="B">a E V</assumption> <conclusion xml:id="C">a f U n V</conclusion> <conclusion xml:id="moon_cheese">made_of(moon,gc)</conclusion> </FMP> </pre> </pre>		<pre> <CMP>If $a \in U$ and $a \in V$, then $a \in U \cap V$ or <phrase index="moon_cheese">the moon is made of green cheese</phrase>. </CMP> <FMP> <assumption xml:id=" A" > $a \in U$</assumption> <assumption xml:id="B">$a \in V$</assumption> <conclusion xml:id="C">$a \in U \cap V$</conclusion> <conclusion xml:id="moon_cheese">$\text{made_of}(\text{moon}, \text{gc})$</conclusion> </FMP> </pre>
kohlhase-omdoc_DIA_0086	134	—	—	—	<pre> <CMP> If <phrase verbalizes="#isaG">\langle G, \circ \rangle is a group</phrase>, then of course <phrase verbalizes="#isaM">it is a monoid</phrase> by construction. </CMP> <FMP> <OMOBJ> <OMA><OMS cd="logic1" name="implies" /> <OMA id="isaG"><OMS cd="algebra" name="group" /> <OMA id="GG"><OMS cd="set" name="pair"> <OMV name="G"/><OMV name="op"/> </OMA> </OMA> <OMA xml:id="isaM"><OMS cd="algebra" name="monoid"/> <OMR href="GG"/> </OMA> </OMA> </OMOBJ> </FMP> </pre>
kohlhase-omdoc_DIA_0087	134	—	—	—	<pre> <OMOBJ> <OMA><OMS cd="set1" name="in" /> <OMS cd="nat" name="zero" /> <OMS cd="nat" name="Nats" /> </OMA> </OMOBJ> </pre>
kohlhase-omdoc_DIA_0088	135	—	—	—	<pre> ...<term cd="nat" name="zero">the number zero</term> is an <term cd="nat" name="Nats">natural number</term>... </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0094	146	—	—	—	<pre> <definition xml:id="exp-def" for="#exp" type="implicit" uniqueness="#exp-unique" existence="#exp-exists"> <FMP>$exp' = exp \wedge exp(0) = 1$</FMP> </definition> 5 <assertion xml:id="exp-unique"> <CMP> There is at most one differentiable function that solves the differential equation in definition <ref type="cite" xref="#exp-def"/>. </CMP> 10 </assertion> <assertion xml:id="exp-exists"> <CMP> The differential equation in <ref type="cite" xref="#exp-def"/> is solvable. </CMP> 15 </assertion> </pre>
kohlhase-omdoc_DIA_0095	147	—	—	—	<pre> <definition xml:id="plus.def" for="#plus" type="inductive" consistency="#s-not-0" exhaustivity="#s-or-0"> <metadata><dc:subject>addition</dc:subject></metadata> <CMP>Addition is defined by recursion on the second argument.</CMP> <requation>$x + 0 \rightsquigarrow x$</requation> <requation>$x + s(y) \rightsquigarrow s(x + y)$</requation> </definition> </pre>
kohlhase-omdoc_DIA_0096	148	—	—	—	<pre> <assertion xml:id="ida.c6s1p4.11" type="lemma"> <CMP> A semigroup has at most one unit.</CMP> <FMP>$\forall S.sgrp(S) \rightarrow \forall x,y.unit(x,S) \wedge unit(y,S) \rightarrow x = y$</FMP> </assertion> </pre>
kohlhase-omdoc_DIA_0097	150	—	—	—	
kohlhase-omdoc_DIA_0098	151	—	—	—	<pre> <type xml:id="double-even.td" system="#POST" theory="adv.int" for="#plus" just-by="#double-even"> <m:math> <m:apply><m:plus/> <m:ci type="integer">X</m:ci> <m:ci type="integer">X</m:ci> </m:math> <m:math> <m:csymbol definitionURL="http://www.mathweb.org/cd/integers/evens"/> </m:math> </type> <assertion xml:id="double-even" type="lemma" theory="adv.int"> <FMP> <m:math> <m:apply><m:forall/> <m:bvar><m:ci xml:id="x13" type="integer">X</m:ci></m:bvar> <m:apply><m:in/> <m:apply><m:plus/> <m:ci definitionURL="x13" type="integer">X</m:ci> <m:ci definitionURL="x13" type="integer">X</m:ci> </m:math> <m:csymbol definitionURL="http://www.mathweb.org/cd/nat/evens"/> </m:math> </FMP> </assertion> </pre>

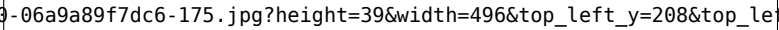
Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0099	153	—	—	—	<pre> <symbol name="strings-over" /> <definition xml:id="strings.def" for="#strings-over">... A* ...</definition> <symbol name="concat" /> <definition xml:id="concat.def" for="#concat">... :: ...</definition> <symbol name="empty-string" /> <definition xml:id="empty-string.def" for="#empty-string">... ε ...</definition> ... <assertion xml:id="string.struct.monoid" type="lemma"> <CMP>(A*, ::, ε) is a monoid.</CMP> <FMP>mon(A*, ::, ε)</FMP> </assertion> ... <example xml:id="mon.ex1" for="#monoid" type="for" assertion="string.struct.monoid"> <CMP>The set of strings with concatenation is a monoid.</CMP> <OMOBJ> <OMA id="nat-strings"> <OMS cd="strings" name="strings" /> <OMS cd="setname1" name="N" /> </OMA> </OMOBJ> <OMOBJ><OMS cd="strings" name="concat" /></OMOBJ> <OMOBJ><OMS cd="strings" name="empty-string" /></OMOBJ> </example> <assertion xml:id="monoid.are.groups" type="false-conjecture"> <CMP>Monoids are groups.</CMP> <FMP>∀S, o, e.mon(S, o, e) → ∃i.group(S, o, e, i)</FMP> </assertion> <example xml:id="mon.ex2" for="#monoids.are.groups" type="against" assertion="strings.isnt.group"> <CMP>The set of strings with concatenation is not a group.</CMP> <OMOBJ><OMR href="#nat-strings" /></OMOBJ> <OMOBJ><OMS cd="strings" name="strings" /></OMOBJ> <OMOBJ><OMS cd="strings" name="concat" /></OMOBJ> <OMOBJ><OMS cd="strings" name="empty-string" /></OMOBJ> </example> <assertion xml:id="strings.isnt.group" type="theorem"> <CMP>(A*, ::, ε) is a monoid, but there is no inverse function for it.</CMP> </assertion> </pre>
kohlhase-omdoc_DIA_0100	154	—	—	—	<pre> <assertion type="theorem" style="display=flow"> <CMP>In a monoid M, the left unit and the right unit coincide,</CMP> </assertion> <definition for="#unit" style="display:flow"> <CMP>we call it the <term role="definiens" name="unit">unit</term> of M</CMP> </definition> </pre>
kohlhase-omdoc_DIA_0101	155	—	—	—	<pre> <assertion xml:id='unit-unique' type="theorem"> <CMP>In a monoid M, the left unit and the right unit coincide, <phrase verbalizes="#unit-def">we call it the unit of M</phrase>.</CMP> </assertion> <symbol name="unit" /> <definition xml:id="unit-def" for="#unit" just-by='#unit-unique'> <CMP>We call the (unique) element of a monoid M that acts as a left and right unit the <term role="definiens" name="unit">unit</term> of M.</CMP> </definition> </pre>
kohlhase-omdoc_DIA_0102	156	—	—	—	<pre> <theory xml:id="a.thy"> <symbol name="aa" /> <theory xml:id="b.thy"> <symbol name="cc" /> <definition xml:id="cc.def" for="#cc" type="simple"> <OMOBJ><OMS cd="a.thy" name="af" /></OMOBJ> </definition> </theory> </theory> </pre>
kohlhase-omdoc_DIA_0103	157	—	—	—	<pre> <theory xml:id="a.thy"><symbol name="aa" /></theory> <theory xml:id="b.thy"> <imports from="#a.thy" type="global" /> <symbol name="cc" /> <definition xml:id="cc.def" for="#cc" type="simple"> <OMOBJ><OMS cd="a.thy" name="af" /></OMOBJ> </definition> </theory> </pre>

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kohlhase-omdoc_DIA_0104	157	—	—	—	<pre> <theory xml:id="A"> <imports from="http://red.com/theories.omdoc#mythy"/> </theory> <theory xml:id="B"> <imports from="http://blue.org/cd/all.omdoc#mythy"/> </theory> <theory xml:id="C"><imports from="#A"/><imports from="#B"/></theory> </pre>
kohlhase-omdoc_DIA_0105	158	—	—	—	<pre> <theory xml:id="semigroup"> <symbol name="set"/><symbol name="op"/> <axiom xml:id="closed">... </axiom><axiom xml:id="assoc">... </axiom> </theory> <theory xml:id="monoid"> <imports from="#semigroup"/> <symbol name="neut"/><symbol name="setstar"/> <axiom xml:id="left-unit"> <CMP>neut is a left unit for op.</CMP><FMP>∀x ∈ set.op(x,neut) = x</FMP> </axiom> <definition xml:id="setstar.def" for="#setstar" type="implicit"> <CMP>.* subtracts the unit from a set </CMP><FMP>∀S.S* = S\{unit}</FMP> </definition> </theory> <theory xml:id="group"> <imports from="#monoid"/> <symbol name="inv"/> <axiom xml:id="left-inv"> <CMP>For every X ∈ set there is an inverse inv(X) wrt. op.</CMP> </axiom> </theory> </pre>
kohlhase-omdoc_DIA_0106	163	—	—	—	<pre> <theory xml:id="Nat"> <adt xml:id="nat-adt"> <metadata> <dc:title>Natural Numbers as an Abstract Data Type.</dc:title> <dc:description>The Peano axiomatization of natural numbers.</dc:description> </metadata> <sortdef name="pos-nats" type="free"> <metadata> <dc:description>The set of positive natural numbers.</dc:description> </metadata> <constructor name="succ"> <metadata><dc:description>The successor function.</dc:description></metadata> <argument> <type><OMOBJ><OMS cd='Nat' name="nats"/></OMOBJ></type> <selector name="pred" total="yes"> <metadata><dc:description>The predecessor function.</dc:description></metadata> </selector> </argument> </constructor> <recognizer name="positive"> <metadata> <dc:description> The recognizer predicate for positive natural numbers. </dc:description> </metadata> </recognizer> </sortdef> </adt> </theory> </pre>
kohlhase-omdoc_DIA_0107	164	—	—	—	<pre> </metadata> </recognizer> </sortdef> <sortdef name="nats" type="free"> <metadata><dc:description>The set of natural numbers</dc:description></metadata> <constructor name="zero"> <metadata><dc:description>The number zero.</dc:description></metadata> </constructor> <insert for="#pos-nats"/> </sortdef> </adt> </theory> </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0108	167	—	—	—	<pre> <assertion xml:id="a1"> <CMP>There are infinitely many prime numbers.</CMP> </assertion> <proof xml:id="p" for="#a1"> <omtext xml:id="intro"> <CMP>We need to prove that the set P of all prime numbers is not finite.</CMP> </omtext> <derive xml:id="d1"> <CMP>We proceed by assuming that P is finite and reaching a contradiction.</CMP> </method> <proof xml:id="p1"> <hypothesis xml:id="h2"><CMP>Let P be finite.</CMP></hypothesis> <derive xml:id="d3"> <CMP>Then $P = \{p_1, \dots, p_n\}$ for some p_i.</CMP> <method><premise xref="#h2"/></method> </derive> <symbol name="q"/> <definition xml:id="d4" for="#q" type="informal"> <CMP>Let $q \stackrel{def}{=} p_1 \cdots p_n + 1$</CMP> </definition> <derive xml:id="d5"> <CMP>Since for each $p_i \in P$ we have $q > p_i$, we conclude $q \notin P$.</CMP> </derive> <omtext xml:id="c6"> <CMP>We prove the absurdity by showing that q is prime:</CMP> </omtext> <derive xml:id="d7"> <CMP>For each $p_i \in P$ we have $q = p_i k + 1$ for some natural number k, so p_i can not divide q.</CMP> <method><premise xref="#d4"/></method> </derive> <derive xml:id="d8"> <CMP>q must be prime as P is the set of all prime numbers.</CMP> <method><premise xref="#d7"/></method> </derive> <derive xml:id="d9"> <CMP>Thus we have contradicted our assumption</CMP> <method><premise xref="#d5"/><premise xref="#d8"/></method> </derive> </proof> </proof> <derive xml:id="d10" type="conclusion"> <CMP>This proves the assertion.</CMP> </derive> </proof> </pre>
kohlhase-omdoc_DIA_0109	171	—	—	—	<pre> <proof xml:id="proof.2.1.2.proof.D2.1" for="#assertion.2.1.2"> ... <derive xml:id="D2.1"> <CMP>By <ref type="cite" xref="#A2"/>, <ref type="cite" xref="#A4"/>, and <ref type="cite" xref="#A5"/> we have $z + (a + (-a)) = (z + a) + (-a)$.</CMP> <FMP>$z + (a + (-a)) = (z + a) + (-a)$</FMP> <method xref="nk-sorts.omdoc#NK-Sorts.forallistar"> <OMOBJ><OMV name="z"/></OMOBJ> <OMOBJ><OMV name="a"/></OMOBJ> <OMOBJ>$-a$</OMOBJ> <premise xref="#A2"/><premise xref="#A4"/><premise xref="#A5"/> </method> </derive> ... </proof> ... <theory xml:id="NK-Sorts"> <metadata> <dc:title>Natural Deduction for Sorted Logic</dc:title> </metadata> <symbol name="forallistar"> <metadata> <dc:description>Repeated Universal Instantiation</dc:description> </metadata> </symbol> <definition xml:id="forallistar.def" for="#forallistar" type="informal"> <CMP>Given n parameters, the inference rule $\forall I^*$ instantiates the first n universal quantifications in the antecedent with them.</CMP> </definition> ... </theory> </pre>

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kohlhase-omdoc_DIA_0110	172	—	—	—	<pre> <assertion xml:id="t1" theory="sets"> <CMP>If $a \in U$ or $a \in V$, then $a \in U \cup V$.</CMP> <FMP> <assumption xml:id="t1_a">$a \in U \vee a \in V$</assumption> <conclusion xml:id="t1_c">$a \in U \cup V$</conclusion> </FMP> </assertion> <proof xml:id="t1_p1" for="#t1" theory="sets"> <omtext xml:id="t1_p1_m1"> <CMP> We prove the assertion by a case analysis.</CMP> </omtext> <derive xml:id="t1_p1_l1"> <CMP>If $a \in U$, then $a \in U \cup V$.</CMP> <FMP> <assumption xml:id="t1_p1_l1_a">$a \in U$</assumption> <conclusion xml:id="t1_p1_l1_c">$a \in U \cup V$</conclusion> </FMP> <method xref="sk.omdoc#SK.by.definition">$\cup$</method> </derive> <derive xml:id="t1_p1_l2"> <CMP>If $a \in V$, then $a \in U \cup V$.</CMP> <FMP> <assumption xml:id="t1_p1_l2_a">$a \in V$</assumption> <conclusion xml:id="t1_p1_l2_c">$a \in U \cup V$</conclusion> </FMP> <method xref="sk.omdoc#SK.by.definition">$\cup$</method> </derive> <derive xml:id="t1_p1_c"> <CMP> We have considered both cases, so we have $a \in U \cup V$.</CMP> </derive> </proof> </pre>
kohlhase-omdoc_DIA_0111	173	—	—	—	<pre> <definition xml:id="union.def" for="#union"> <OMOBJ>$\forall P, Q, x. x \in P \cup Q \Leftrightarrow x \in P \vee x \in Q$</OMOBJ> </definition> <proof xml:id="t1_p1_l1.exp" for="#t1_p1_l1"> <derive xml:id="t1_p1_l1.d1"> <FMP> <assumption xml:id="t1_p1_l1.d1_a">$a \in U$</assumption> <conclusion xml:id="t1_p1_l1.d1_c">$a \in U$</conclusion> </FMP> <method xref="sk.omdoc#SK.axiom"/> </derive> <derive xml:id="t1_p1_l1.l1.d2"> <FMP> <assumption xml:id="t1_p1_l1.d2_a">$a \in U$</assumption> <conclusion xml:id="t1_p1_l1.d2_c">$a \in U \vee a \in V$</conclusion> </FMP> <method xref="sk.omdoc#SK.orR"><premise xref="#t1_p1_l1.d1"/></method> </derive> <derive xml:id="t1_p1_l1.d3"> <FMP> <assumption xml:id="t1_p1_l1.d3_a">$a \in U \vee a \in V$</assumption> <conclusion xml:id="t1_p1_l1.d3_c">$a \in U \cup V$</conclusion> </FMP> <method xref="sk.omdoc#SK.definition-rl">U, V, a <premise xref="#unif.def"/> </method> </derive> <derive xml:id="t1_p1_l1.d4"> <FMP> <assumption xml:id="t1_p1_l1.d3_a">$a \in U$</assumption> <conclusion xml:id="t1_p1_l1.d3_c">$a \in U \cup V$</conclusion> </FMP> <method xref="sk.omdoc#SK.cut"> <premise xref="#t1_p1_l1.d2"/> <premise xref="#t1_p1_l1.d3"/> </method> </derive> </proof> </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0112	174	—	—	—	<assertion xml:id="a1"> <CMP>There are infinitely many prime numbers.</CMP> </assertion> <proof for="#a1"> <omtext xml:id="c0"> <CMP>We need to prove that the set P of all prime numbers is not finite.</CMP> </omtext> <derive xml:id="d1"> <CMP> We proceed by assuming that P is finite and reaching a contradiction.</CMP> <method xref="nk.omdoc#NK.by-contradiction"> <proof> <hypothesis xml:id="h2"><CMP>Let P be finite.</CMP></hypothesis> <derive xml:id="d3"><CMP>Then $P = \{p_1, \dots, p_n\}$ for some n</CMP></derive> <symbol name="q"/> <definition xml:id="d4" for="#q" type="informal">

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0113	175	—	<pre> \begin{proof} \text{Let } q \stackrel{\text{def}}{=} p_1 \cdots p_n + 1. \text{For each } p_i \in P \text{ we have } q > p_i. \text{We show absurdity by showing that } q \text{ is prime.} \text{Assume } q \text{ is not prime. Then } q = p \cdot r \text{ for some } p, r \in P. \text{Then } p \mid p_1 \cdots p_n + 1. \text{But } p \mid p_1 \cdots p_n \text{ since } p \in P. \text{Hence } p \mid 1, \text{ which is a contradiction.} \text{Therefore } q \text{ is prime.} \end{proof} </pre>		<pre> <CMP>Let $q \stackrel{\text{def}}{=} p_1 \cdots p_n + 1$</CMP> </definition> <derive xml:id="d5a"> <CMP>For each $p_i \in P$ we have $q > p_i$</CMP> <method xref="#Trivial"><premise xref="#d4"/></method> </derive> <derive xml:id="d5b"> <CMP>$q \notin P$</CMP> <method xref="#Trivial"><premise xref="#d5"/></method> </derive> <derive xml:id="d6"> <CMP>We show absurdity by showing that q is prime</CMP> <FMP>$\perp$</FMP> <method xref="#Contradiction"> <premise xref="#d5b"/> </method> <proof> <derive xml:id="d7a"> <CMP> For each $p_i \in P$ we have $q = p_i k + 1$ for a given natural number k. </CMP> <method xref="#By_Definition"><premise xref="#d1"/></method> </derive> <derive xml:id="d7b"> <CMP>Each $p_i \in P$ does not divide q</CMP> </derive> <derive xml:id="d8"> <CMP>q is prime</CMP> <method xref="#Trivial"> <premise xref="#h2"/> <premise xref="#p4"/> </method> </derive> </proof> </method> </derive> </proof> </method> </derive> </proof> </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0114	176	—	—	—	$\frac{\frac{[A \wedge B]}{B} \wedge E_r \quad \frac{[A \wedge B]}{A} \wedge E_l}{B \wedge A} \wedge I$ $\frac{B \wedge A}{A \wedge B \Rightarrow B \wedge A} \Rightarrow I$ <pre> <proofobject xml:id="ac.p" for="#and-comm"> <metadata> <dc:description> Assuming $A \wedge B$ we have B and A from which we can derive $B \wedge A$. </dc:description> </metadata> <OMOBJ> <OMBIND id="andcom.pf"> <OMS cd="PL0ND" name="impliesI"> <OMBVAR> <OMATTR> <OMATP> <OMS cd="PL0ND" name="type"> $A \wedge B$ </OMATP> <OMV name="X"> </OMATP> </OMBVAR> </OMATTR> <OMA> <OMS cd="PL0ND" name="andI"> <OMA> <OMS cd="PL0ND" name="andEr"> <OMV name="X"> </OMV> </OMS> </OMA> <OMS cd="PL0ND" name="andEl"> <OMV name="X"> </OMV> </OMS> </OMA> </OMBIND> </OMOBJ> </proofobject> </pre> The schema on the left shows the proof as a natural deduction proof tree, the OMDoc representation gives the proof object as a λ term. This term would be written as the following term in traditional (mathematical) notation: $\Rightarrow I(\lambda X : A \wedge B. \wedge I(\wedge E_r(X), \wedge E_l(X)))$
kohlhase-omdoc_DIA_0115	177	—	—	—	<pre> <theory xml:id="PL0ND"> <metadata> <dc:description>The Natural Deduction Calculus for Propositional Logic</dc:description> </metadata> ... <symbol name="andI"> <metadata><dc:subject>Conjunction Introduction</dc:subject></metadata> <type system="prop-as-types">$A \rightarrow B \rightarrow (A \wedge B)$</type> </symbol> <definition xml:id="andI.def" for="#andi"> <CMP>Conjunction introduction, if we can derive A and B, then we can conclude $A \wedge B$.</CMP> </definition> ... </theory> </pre>
kohlhase-omdoc_DIA_0116	177	—	—	—	<pre> <symbol name="andcom"> <metadata><dc:description>Commutativity for $\wedge$</dc:description></metadata> <type system="prop-as-types">$(A \wedge B) \rightarrow (B \wedge A)$</type> </symbol> <definition xml:id="andcom.def" for="#andcom" type="simple"> <OMOBJ><OMR href="#andcom.pf"/></OMOBJ> </definition> </pre>
kohlhase-omdoc_DIA_0117	179	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0118	180	—	—	—	
kohlhase-omdoc_DIA_0119	180	—	—	—	<pre> <theory xml:id="ring"> <symbol name="times"/><symbol name="one"/> <imports xml:id="add.import" from="#group" type="global"/> <imports xml:id="mult.import" from="#monoid" type="global"> <morphism> <requation> <OMOBJ><OMS cd="monoid" name="set"/></OMOBJ> <OMOBJ> <OMA><OMS cd="monoid" name="setstar"/> <OMS cd="semigroup" name="set"/> </OMA> </OMOBJ> </requation> </theory> <requation> <OMOBJ><OMS cd="monoid" name="op"/></OMOBJ> </requation> </pre>
kohlhase-omdoc_DIA_0120	184	—	<pre> ![[https://cdn.mathpix.com/cropped/1b348de5-ed54-4a9a-aba0-6ba4e2e2e2e2.jpg?height=34&width=1024&top_left_y=213&top_left_x=1024]] \` <theory-inclusion xml:id="grp-conv-grp" from="#group" to="#group"> <morphism><requation> \((\square / 木) 貝 \ <obligation assertion="#conv.closed" induced-by="#closed.ax"/> <obligation assertion="#conv.assoc" induced-by="#assoc.ax"/> <obligation assertion="#left.unit" induced-by="#unit.ax"/> <obligation assertion="#conv.inv" induced-by="#inv.ax"/> </theory-inclusion> \` </pre>		<pre> <assertion xml:id="conv.inv" theory="group">\forall x,y \in M.x \circ x^{-1} = e</assertion> <theory-inclusion xml:id="grp-conv-grp" from="#group" to="#group"> <morphism><requation>X \circ Y \rightsquigarrow Y \circ X</requation></morphism> <obligation assertion="#conv.closed" induced-by="#closed.ax"/> <obligation assertion="#conv.assoc" induced-by="#assoc.ax"/> <obligation assertion="#left.unit" induced-by="#unit.ax"/> <obligation assertion="#conv.inv" induced-by="#inv.ax"/> </theory-inclusion> </pre>
kohlhase-omdoc_DIA_0121	185	—	—	—	
kohlhase-omdoc_DIA_0122	185	—	—	—	<pre> <assertion xml:id="foo" type="theorem">...</assertion> <proof xml:id="foo.pf" for="#foo">...</proof> <assertion xml:id="target" induced-by="#foo" induced-via="#grp-conv-grp"> ... </assertion> </pre>

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kohlhase-omdoc_DIA_0123	188	—	—	—	
kohlhase-omdoc_DIA_0124	188	—	—	—	
kohlhase-omdoc_DIA_0125	189	—	—	—	
kohlhase-omdoc_DIA_0126	190	—	—	—	<pre> <theory xml:id="t1">...</theory> <theory xml:id="t2">...</theory> <theory xml:id="a1"> <imports xml:id="ima1" from="#t1"/> <axiom xml:id="axa11">...</axiom> <axiom xml:id="axa12">...</axiom> </theory> <theory xml:id="a2"> <imports xml:id="ima2" from="#t1"/> <imports xml:id="im2a2" from="#t2"/> <axiom xml:id="axa21">...</axiom> </theory> <theory xml:id="c1"> <imports xml:id="im1c1" from="#a1"/> <imports xml:id="im2c1" from="#b1"/> <axiom xml:id="axc11">...</axiom> </theory> <theory xml:id="b1"> <imports xml:id="imb1" from="#t2"/> <axiom xml:id="axb11">...</axiom> </theory> <theory xml:id="b2"> <imports xml:id="imb2" from="#t2"/> <axiom xml:id="axb21">...</axiom> </theory> <theory xml:id="c2"> <imports xml:id="im1c2" from="#a2"/> <imports xml:id="im2c2" from="#b2"/> <axiom xml:id="axc21">...</axiom> </theory> </pre>
kohlhase-omdoc_DIA_0127	190	—	—	—	<pre> <axiom-inclusion xml:id="aia" from="#a1" to="#a2"> <obligation induced-by="#axa11" assertion="#th-axa11"/> <obligation induced-by="#axa12" assertion="#th-axa12"/> </axiom-inclusion> <axiom-inclusion xml:id="bib" from="#b1" to="#b2"> <obligation induced-by="#axb11" assertion="#th-axb11"/> </axiom-inclusion> <axiom-inclusion xml:id="cic" from="#c1" to="#c2"> <obligation induced-by="#axc11" assertion="#th-axc11"/> </axiom-inclusion> </pre>
kohlhase-omdoc_DIA_0128	191	—	—	—	<pre> <axiom-inclusion xml:id="t1ic" from="#t1" to="#c2"> <path-just local="#im1a2" globals="#im1c2"/> </axiom-inclusion> <axiom-inclusion xml:id="t2ic" from="#t2" to="#c2"> <path-just local="#imb2" globals="#im2c2"/> </axiom-inclusion> <axiom-inclusion xml:id="aic" from="#a1" to="#c2"> <path-just local="#aia" globals="#im1c2"/> </axiom-inclusion> <axiom-inclusion xml:id="bic" from="#b1" to="#c2"> <path-just local="#bib" globals="#im2c2"/> </axiom-inclusion> </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0129	191	—	—	—	<pre> <theory-inclusion xml:id="tcic" from="#c1" to="#c2"> <decomposition links="#t1ic #t2ic #aic #bic #cic"/> </theory-inclusion> </pre>
kohlhase-omdoc_DIA_0130	194	—	—	—	<pre> <CMP> I want to mark <phrase xml:id="w1" class="important">this important text</phrase> as special.<phrase class="linebreak"/> </CMP> <omstyle element="omdoc:phrase" class="important"> <style format='html pmml'><element name="em"><recurse/></element></style> <xslt format='TeX' xmlns:xsl="http://www.w3.org/1999/XSL/Transform"> <xsl:text>\emph{</xsl:text> <xsl:apply-templates/> <xsl:text>}</xsl:text> </xslt> </omstyle> <omstyle element="omdoc:phrase" class="linebreak"> <style format='html pmml'><element name="br"/></style> <style format='TeX'><text>\par\noindent</text></style> </omstyle> </pre>
kohlhase-omdoc_DIA_0131	197	—	—	—	<pre> <xsl:template match="OMBIND[OMS[position()=1 and @name='forall' and @cd='quant1']]"> <xsl:text>∀</xsl:text> <xsl:for-each select="OMBVAR"> <xsl:apply-templates/> <xsl:if test="position()! = last()">,</xsl:if> </xsl:for-each> <xsl:text>.</xsl:text> <xsl:apply-templates select="*[3]"> </xsl:template> </pre>
kohlhase-omdoc_DIA_0132	197	—	—	—	<pre> <OMBIND> <OMS cd="quant1" name="forall"/> <OMBVAR><OMV name="P"/><OMV name="Q"/></OMBVAR> <OMA> <OMS cd="logic1" name="implies"/> <OMA><OMS cd="logic1" name="or"/> <OMV name="P"/> <OMV name="Q"/> </OMA> <OMA><OMS cd="logic1" name="or"/> <OMV name="Q"/> <OMV name="P"/> </OMA> </OMBIND> </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0133	198	—	—	—	<pre> <presentation for="#quant1.forall" role="binding"> <CMP>We write <OMOBJ> <OMBIND><OMS cd="quant1" name="forall" /> <OMBVAR><OMV name="X"></OMBVAR> <OMV name="A" /> </OMBIND> </OMOBJ> for the phrase "A holds for all X". </CMP> <xslt format="default" xmlns:xsl="http://www.w3.org/1999/XSL/Transform"> <xsl:text>∀</xsl:text> <xsl:for-each select="OMBVAR"> <xsl:apply-templates/> <xsl:if test="position()=last()"></xsl:if> </xsl:for-each> <xsl:text>.</xsl:text> <xsl:apply-templates select="*[3]" /> </xslt> <style format="html"> <text>&#8704;</text> <map select="OMBVAR/*"> <separator><text>,</text></separator> <recurse/> </map> <text>.</text> <recurse select="*[3]" /> </style> <style format="pmml"> <element crid="." name="mrow" ns="http://www.w3.org/1998/Math/MathML"> <element crid="*[1]" cr="yes" name="mo"><text>&#8704;</text></element> <element name="mrow" crid="*[2]"> <map select="OMBVAR/*"> <separator> <element name="mo" cr="yes"> <attribute name="separator"><text>true</text></attribute> <text>,</text> </element> </separator> <recurse/> </map> </element> <recurse select="*[3]" /> </element> </style> </presentation> </pre>
kohlhase-omdoc_DIA_0134	200	—	—	—	<pre> <OMA> <OMS cd="arith1" name="power" /> <OMA> <OMS cd="arith1" name="plus" /> <OMV name="x" /> <OMV name="x" /> <OMV name="y" /> </OMA> <OMI>2</OMI> </OMA> <OMA> <OMS cd="arith1" name="plus" /> <OMV name="x" /> <OMV name="y" /> </OMA> <OMI>2</OMI> </OMA> </pre>
kohlhase-omdoc_DIA_0135	202	—	<pre> \begin{lstlisting}[mathescape=true] <presentation for=" #forall" role="binding" separator = ". " \(>\) <use format="TeX"> forall</use> <use format="html">&#8704;</use> </presentation> \end{lstlisting} </pre>	(not rendered)	<pre> <presentation for="#forall" role="binding" separator="." "> <use format="TeX">\forall</use> <use format="html">&#8704;</use> </presentation> </pre>
kohlhase-omdoc_DIA_0136	202	—	<pre> \begin{lstlisting}[mathescape=true] <OMBIND> <OMS cd="quant1" name="forall"/> <OMBVAR> <OMV name="X"/> </OMBVAR> <OMS cd="logic1" name="true"/> </OMBIND> \end{lstlisting} </pre>	(not rendered)	<pre> <OMBIND> <OMS cd="quant1" name="forall" /> <OMBVAR> <OMV name="X" /> </OMBVAR> <OMS cd="logic1" name="true" /> </OMBIND> </pre>

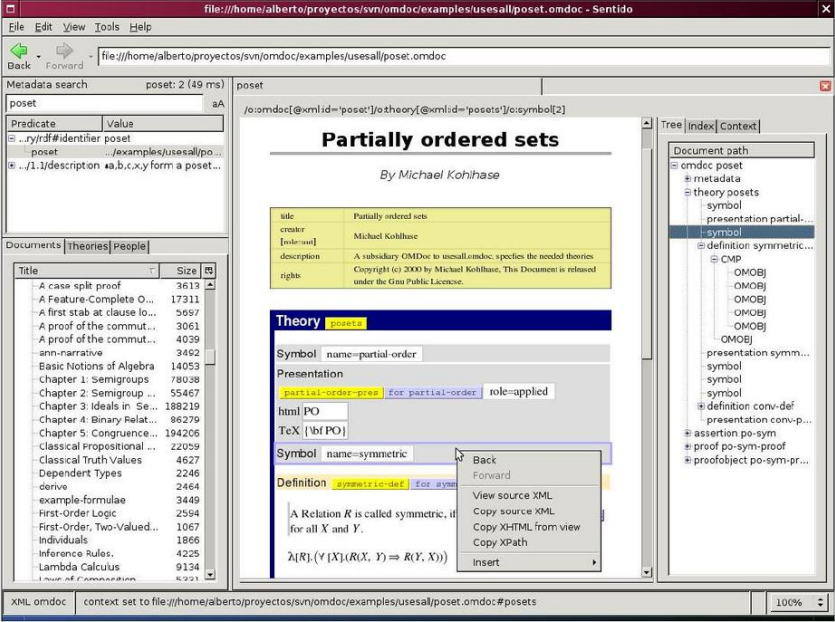
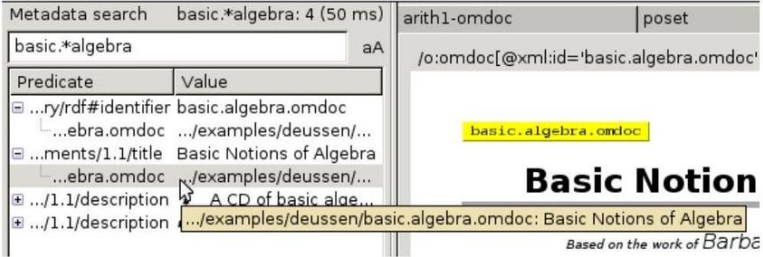
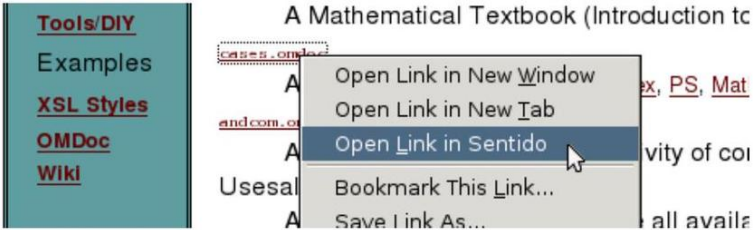
Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0144	205	—	—	—	<pre> <OMOBJ xmlns="http://www.openmath.org/OpenMath"> <OMBIND> <OMS cd="quant1" name="forall"/> <OMBVAR> <OMATTR> <OMATP> <OMS cd="omdoc" name="notation"/> <OMFOREIGN encoding="application/omdoc+xml"> <presentation xmlns="http://www.mathweb.org/omdoc" for="#R" role="applied" precedence="500" fixity="infix"> <use format="TeX">\prec</use> <use format="pmml html">&#x022DE;</use> </presentation> <presentation xmlns="http://www.mathweb.org/omdoc" for="#R"> <use format="TeX">{\prec{}}</use> <use format="pmml html">&#x022DE;</use> </presentation> </OMFOREIGN> </OMATP> </OMATTR> <OMV name="R"/> </OMBVAR> <OMA><OMV name="R"/><OMV name="X"/><OMV name="X"/></OMA> </OMBIND> </OMOBJ> </pre>
kohlhase-omdoc_DIA_0145	208	—	—	—	<pre> <assertion xml:id="ALGX0"> <CMP>If a, b, c, d are numbers, then we have $a + b(c + d) = a + bc + bd$.</CMP> </assertion> <private xml:id="alg-expr-1" pto="Analytica" reformulates="ALGX0"> <data format="mathematica-5.0"> <![CDATA[SIMPLIFYRULES[a_ + b_*(c_ + d_) := a + b*c + b*d /; NumberQ[b]]]> </data> </private> </pre>
kohlhase-omdoc_DIA_0146	209	—	—	—	<pre> <private xml:id="legacy"> <metadata> <dc:title>A fragment of Bourbaki's Algebra</dc:title> <dc:creator role="trl">Michael Kohlhase</dc:creator> <dc:date action="created">2002-01-03T0703</dc:date> <dc:description>A fragment of Bourbaki's Algebra</dc:description> <dc:source>Nicolas Bourbaki, Algebra, Springer Verlag 1974</dc:source> <dc:type>Text</dc:type> </metadata> <data format="application/x-latex" href="legacy.tex"/> <data format="image/jpeg" href="legacy.jpeg"/> <data format="application/postscript" href="legacy.ps"/> <data format="application/pdf" href="legacy.pdf"/> </private> </pre>
kohlhase-omdoc_DIA_0147	209	—	—	—	<pre> <effect><CMP>None.</CMP></effect> </pre>
kohlhase-omdoc_DIA_0148	209	—	—	—	<pre> <code xml:id="callMint" requires="org.riaca.cas"> <metadata> <dc:description> The multiple integrator applet. It puts up a user interface , queries the user for a function , which it then integrates by calling one of several computer algebra systems. </dc:description> </metadata> <data format="application/x-java-applet"> <![CDATA[... <i>the callMint code goes here</i> ...]]> </data> <input><CMP>None: the applet handles input itself.</CMP></input> <output><CMP>The result of the integration.</CMP></output> <effect><CMP>None.</CMP></effect> </code> </pre>
kohlhase-omdoc_DIA_0149	212	—	—	—	<pre> <omlet data="#legacy" show="embed">A Fragment of Bourbaki's Algebra</omlet> </pre>

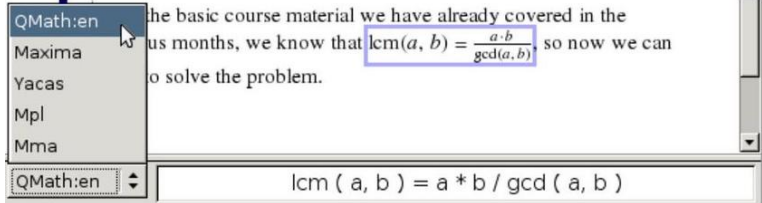
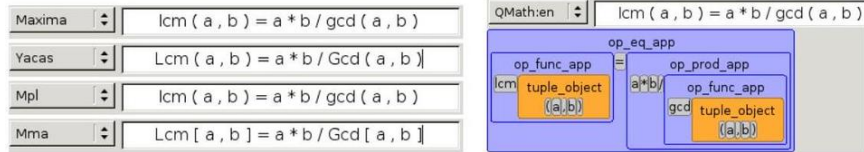
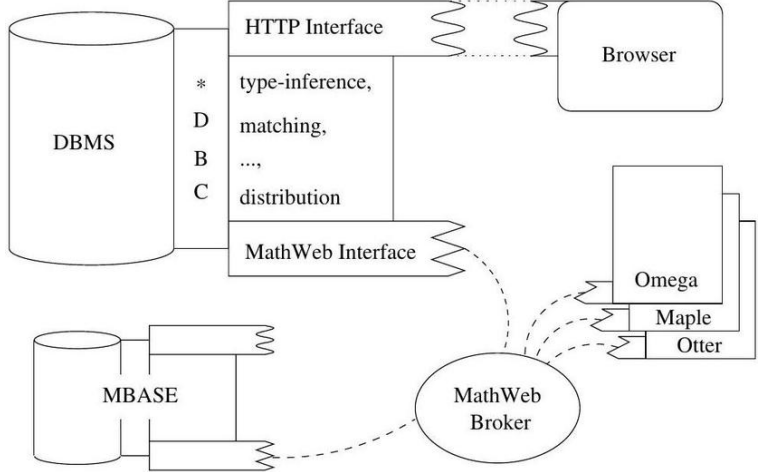
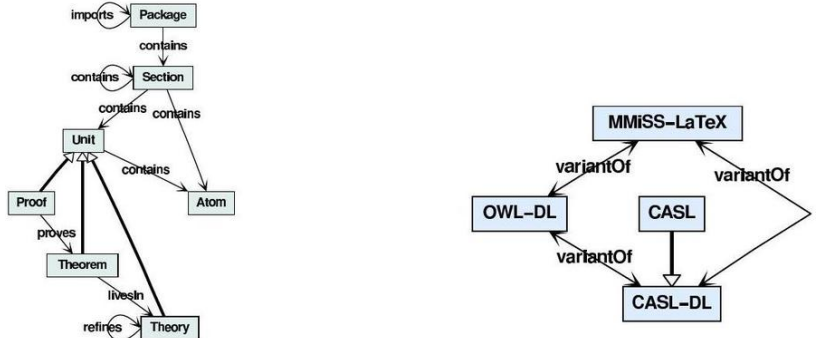
Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0150	212	—	—	—	<pre> <omtext xml:id="monp_1"> <CMP> <p>Let practice integration!</p> <p><omlet data="#callMint" action="execute" style="width:320;height:200"> No plug-in found for callMint! </omlet></p> </CMP> </omtext> </pre>
kohlhase-omdoc_DIA_0151	212	—	—	—	<pre> <CMP> Let us prove it interactively: <omlet data="#sendtoTP" action="display"> <param name="timeout" value="30" valuetype="data"/> <param name="performative" value="prove"/> <param name="problem" value="#ALGX0" valuetype="object"/> <param name="description" value="http://example.org/prob17.html" valuetype="ref"/> <param name="instance"> <OMOBJ> <OMA><OMS name="root" cd="arith1"/> <OMI>3</OMI><OMI>3</OMI> </OMA> </OMOBJ> </param> Sorry, no theorem prover available! </omlet> </CMP> </pre>
kohlhase-omdoc_DIA_0152	215	—	—	—	<pre> <exercise xml:id="TeXBook-18-22"> <CMP> <p>Sometimes the condition that defines a set is given as a fairly long English description ; for example consider ‘{p p and p+2 are prime}’. An hbox would do the job:</p> <p style="display:block;font-family:fixed"> \$\{\,p\,\mid\,\hbox{\$p\$ and \$p+2\$ are prime}\,\,}\$ </p> <p>but a long formula like this is troublesome in a paragraph, since an hbox cannot be broken between lines, and since the glue inside the <phrase style="font-family:fixed">\hbox</phrase> does not vary with the inter-word glue in the line that contains it . Explain how the given formula could be typeset with line breaks.</p> </CMP> <hint> <CMP>Go back and forth between math mode and horizontal mode.</CMP> </hint> <solution> <CMP> <phrase style="font-family:fixed"> \$\{\,p\,\mid\,p\$ and \$p+2\$ are prime\$\,\,}\$ </phrase>, assuming that <phrase style="font-family:fixed">\mathsurround</phrase> is zero. The more difficult alternative ‘<phrase style="font-family:fixed"> \$\{\,p\,\mid\,p\,\{\rm and\}\,p+2\rm\ are\ prime\,\,}\$</phrase>’ is not a solution , because line breaks do not occur at <phrase style="font-family:fixed">_</phrase> (or at glue of any kin) within math formulas. Of course it may be best to display a formula like this , instead of breaking it between lines. </CMP> </solution> </exercise> </pre>
kohlhase-omdoc_DIA_0153	216	—	—	—	<pre> <exercise for="#ida.c6s1p4.l1" xml:id="ida.c6s1p4.mc1"> <CMP> What is the unit element of the semi-group Q with operation $a * b = 3ab$? </CMP> <mc> <choice><FMP><OMOBJ><OMI>1</OMI></OMOBJ></FMP></choice> <answer verdict="false"><CMP>No, $1 * 1 = 3$ and not 1</CMP></answer> </mc> <mc> <choice><CMP>$1/3$</CMP></choice> <answer verdict="true"></answer> </mc> <mc> <choice><CMP>It has no unit.</CMP></choice> <answer verdict="false"><CMP>No, try another answer</CMP></answer> </mc> </exercise> </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0154	219	—	—	—	<pre> <definition xml:id="comm-def" for="#comm"> <CMP xml:lang="en"> An operation <OMOBJ id="op"><OMV name="op"/></OMOBJ> is called commutative, iff <OMOBJ id="comm1"> <OMA><OMS cd="relation1" name="eq"/> <OMA><OMV name="op"/><OMV name="X"/><OMV name="Y"/></OMA> <OMA><OMV name="op"/><OMV name="Y"/><OMV name="X"/></OMA> </OMA> </OMOBJ> for all <OMOBJ id="x"><OMV name="X"/></OMOBJ> and <OMOBJ id="y"><OMV name="Y"/></OMOBJ>. </CMP> <CMP xml:lang="de"> Eine Operation <OMOBJ><OMR href="#op"/></OMOBJ> heißt kommutativ, falls <OMOBJ><OMR href="#comm1"/></OMOBJ> für alle <OMOBJ><OMR href="#x"/></OMOBJ> und <OMOBJ><OMR href="#y"/></OMOBJ>. </CMP> </definition> </pre>
kohlhase-omdoc_DIA_0155	219	—	—	—	<pre> <definition for="#comm" xml:id="comm-def"> ... <CMP xml:lang='de'> <!-- Note the unabbreviated empty element --> Eine Operation <OMOBJ><OMR href="#op"/></OMOBJ> heißt kommutativ, falls <OMOBJ><OMR href='comm1'/></OMOBJ> für alle <OMOBJ><OMR href="#x"/></OMOBJ> und <OMOBJ><OMR href='y'/></OMOBJ>. </CMP> </definition> </pre>
kohlhase-omdoc_DIA_0156	220	—	—	—	<pre> <definition xml:id="comm-def" for="#comm"> <CMP xml:lang="de">Eine Operation <OMOBJ><OMR href="#op"/></OMOBJ> heißt kommutativ, falls <OMOBJ id="comm1"> <OMA><OMS cd="relation1" name="eq"/> <OMA><OMV name="op"/><OMV name="X"/><OMV name="Y"/></OMA> <OMA><OMV name="op"/><OMV name="Y"/><OMV name="X"/></OMA> </OMA> </OMOBJ> für alle <OMOBJ><OMR href="#x"/></OMOBJ> und <OMOBJ><OMR href="#y"/></OMOBJ>. </CMP> <CMP xml:lang="en"> An operation <OMOBJ id="op"><OMV name="op"/></OMOBJ> is called commutative, iff <OMOBJ><OMR href="#comm1"/></OMOBJ> for all <OMOBJ id="x"><OMV name="X"/></OMOBJ> and <OMOBJ id="y"><OMV name="Y"/></OMOBJ>. </CMP> </definition> </pre>
kohlhase-omdoc_DIA_0157	222	—	—	—	
kohlhase-omdoc_DIA_0158	227	—	—	—	<pre> svn co https://svn.mathweb.org/repos/mathweb.org/branches/omdoc-1.2 </pre>
kohlhase-omdoc_DIA_0159	231	—	—	—	<pre> <!ENTITY % assertiontype "theorem proposition lemma %otherassertiontype;"> </pre>

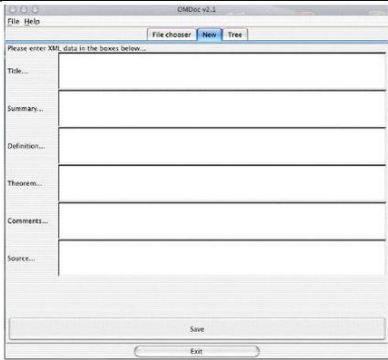
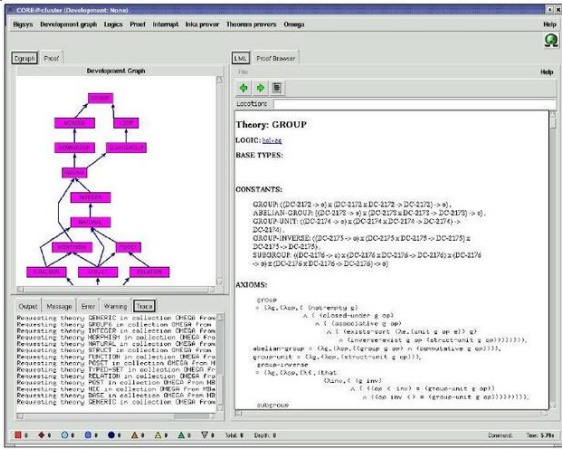
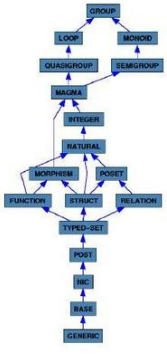
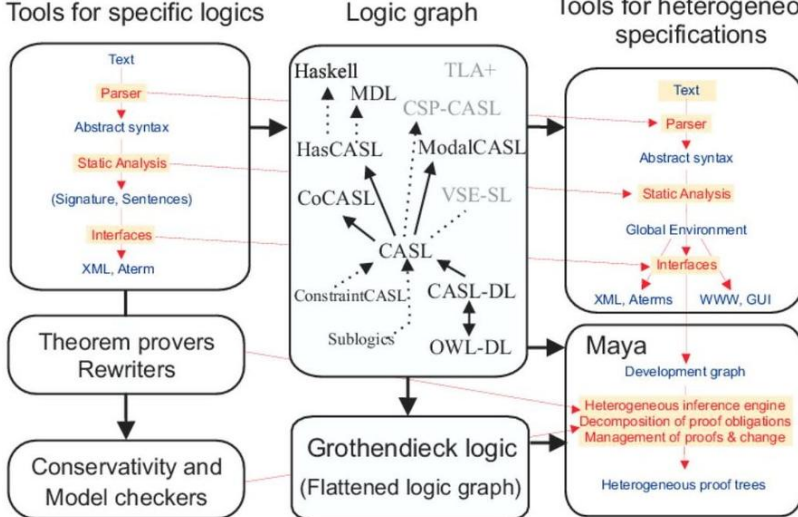
Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0160	231	—	—	—	<pre> <!DOCTYPE omdoc PUBLIC "-//OMDoc//DTD OMDoc V1.2//EN" "http://www.mathweb.org/omdoc/omdoc.dtd" [<!ENTITY % otherassertiontype "observation">]> ... <assertion type="observation">...</assertion> ... </pre>
kohlhase-omdoc_DIA_0161	231	—	—	—	<pre> <?xml version="1.0" standalone="true"?> <omdoc xml:id="something" version="1.2" xmlns="http://www.mathweb.org/omdoc"> ... </omdoc> </pre>
kohlhase-omdoc_DIA_0162	233	—	—	—	<pre> <omdoc:omdoc xmlns:omdoc="http://www.mathweb.org/omdoc"> ... </omdoc:omdoc> </pre>
kohlhase-omdoc_DIA_0163	233	—	—	—	<pre> <!DOCTYPE omdoc:omdoc PUBLIC "-//OMDoc//DTD OMDoc V1.2//EN" "http://www.mathweb.org/omdoc/dtd/omdoc.dtd" [<!ENTITY % NS.prefix "INCLUDE"><!ENTITY % omdoc.prefix "omdoc">]> </pre>
kohlhase-omdoc_DIA_0164	235	—	—	—	<pre> <?xml version="1.0"?> <omdoc xml:id="example.with.schema" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="http://www.mathweb.org/omdoc http://www.mathweb.org/omdoc/xsd/omdoc.xsd"> ... </omdoc> </pre>
kohlhase-omdoc_DIA_0165	238	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0166	240	—	—	—	<symbol name="sigmatype"> <metadata> <dc:description> The dependent function type constructor is a binding operator. The source type is the type of the bound variable X, the target type is represented in the body. </dc:description> </metadata> </symbol> <presentation xml:id="pr-sigmatype" for="#sigmatype" role="binding"> <style format="pvs"> <text></text> <recurse select="*[2]/*"/><text> -&gt; </text><recurse select="*[3]"/> <text></text> </style> <style format="nuprl"> <recurse select="*[2]/*"/><text> -&gt; </text><recurse select="*[3]"/> </style> </presentation>

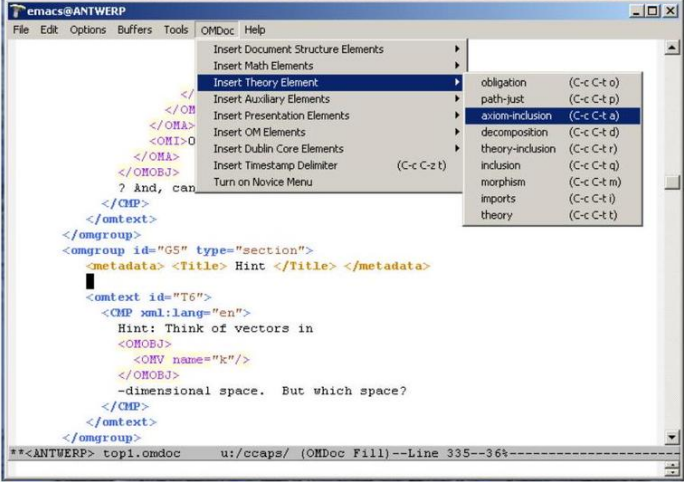
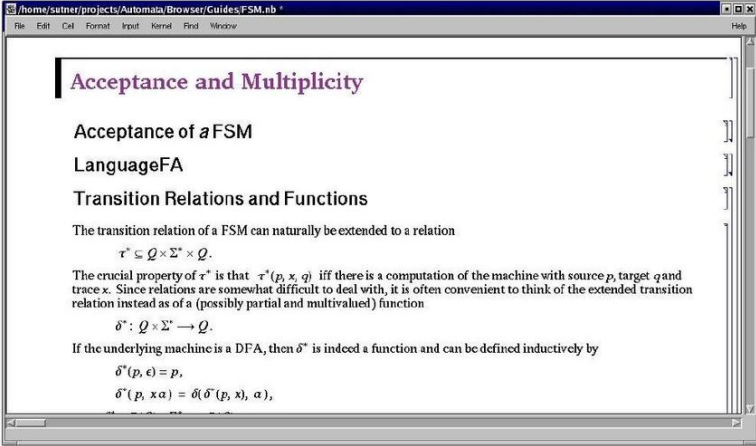
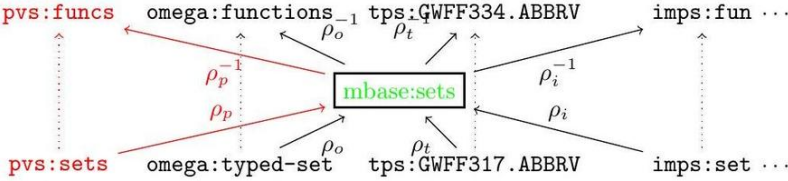
Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0168	247	—	—	—	<pre><?xml version='1.0' encoding='UTF-8' standalone='no'?> <!DOCTYPE omdoc PUBLIC "-//OMDoc/DTD OMDoc V1.2//EN" "-//..../..../ dtd/omdoc.dtd"> <?QMath :en Context: "Mathematics/Arithmetic" ?> <omdoc xmlns='http://www.mathweb.org/omdoc' version='1.2' xmlns:dc='http://purl.org/dc/elements/1.1/'> <metadata> <dc:language>en</dc:language> <dc:title>Diary</dc:title> <dc:creator role='aut'>W. Smith</dc:creator> <dc:date>1984-04-04T18:43:00</dc:date> </metadata> <omtext> <theory xml:id='thoughtcrime'> <omtext> <metadata><dc:title>Down with Big Brother</dc:title></metadata> <CMP> Freedom is the freedom to say $\\$2+2=4\\$. If that is granted, all else follows. </CMP> </omtext> </theory> </omdoc></pre>
kohlhase-omdoc_DIA_0169	248	—	—	—	
kohlhase-omdoc_DIA_0170	249	—	—	—	
kohlhase-omdoc_DIA_0171	250	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0172	250	—	—	—	
kohlhase-omdoc_DIA_0173	251	—	—	—	
kohlhase-omdoc_DIA_0174	254	—	—	—	
kohlhase-omdoc_DIA_0175	257	—	—	—	<pre> <math xmlns="http://www.w3.org/1998/Math/MathML" xmlns:mq="http://mathweb.org/MathQuery"> <apply><int/> <domainofapplication mq:generic="domain"/> <bvar> <ci mq:generic="time"/> </bvar> <apply><power/> <apply><ci mq:generic="fun"></ci><ci mq:generic="time"/></apply> <cn>2</cn> </apply> </math> </pre>
kohlhase-omdoc_DIA_0176	260	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0177	265	—	—	—	The diagram illustrates the interaction between a Document Server and Mathematical WEB Services. The Document Server is enclosed in a green dashed box and contains three components: Static Context Manager (Theory Repository), Dynamic Context Manager (Scopes-Variables Tree), and Document Manager (Presentation Oriented Documents). The Static Context Manager sends a 'Request' to the Document Manager, which responds with a 'Response'. The Dynamic Context Manager sends a 'Request' to the Document Manager, which responds with a 'Response'. The Document Manager sends a 'Request' to the Mathematical WEB Services, which responds with a 'Response'. The Mathematical WEB Services are represented by a dashed box containing four circles labeled CAS (Maple), CAS (Mathematica), CAS (Gap), and TP (Coq). A Client (represented by a computer icon) sends a 'Request' to the Document Manager, which responds with a 'Response'.
kohlhase-omdoc_DIA_0178	268	—	—	—	This diagram shows a complex system architecture for a learning environment. It includes several interconnected components: CAS (Computer Algebra System), exercise repository, MBase (Mathematical Base), mediator, content manager, presentation system, search index, tutorial component, next-best suggestor, tutorial course generator, learner model, domain reasoner, dialogue manager, assessment tool, service relays, ActiveMath front end, controllers (book, search, Tutorial comp., open learner-model, exercise), and views (dialogue console, browser windows, concept-map tool, open learner model, assembling tool, browser windows). The diagram uses solid arrows for 'content flow' and dashed arrows for 'web-service request'. The client interacts with the system through the dialogue console, browser windows, concept-map tool, open learner model, and assembling tool.
kohlhase-omdoc_DIA_0179	277	—	—	—	This diagram illustrates the relationships between different mathematical concepts. A central 'statement' box is connected to 'definition', 'assertion', 'proof', and 'example' boxes. 'definition' is connected to 'assertion' via 'is a' and 'proves' relationships. 'assertion' is connected to 'proof' via 'proves' and 'exemplifies' relationships. 'proof' is connected to 'example' via 'exemplifies' and 'proves' relationships. 'statement' is connected to 'theory' via 'context for' and 'imports' relationships. 'theory' is connected to 'statement' via 'imports' and 'context for' relationships.

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0180	278	—	—	—	depends on: - group - monoid Show these? No, thanks! Yes, explain Yes, explore Later
kohlhase-omdoc_DIA_0181	281	—	—	—	
kohlhase-omdoc_DIA_0182	284	—	—	—	 
kohlhase-omdoc_DIA_0183	289	—	—	—	Tools for specific logics Logic graph Tools for heterogeneous specifications 

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0184	290	—	—	—	
kohlhase-omdoc_DIA_0185	292	—	—	—	
kohlhase-omdoc_DIA_0186	293	—	—	—	
kohlhase-omdoc_DIA_0187	298	—	—	—	$\backslash\begin{module}[id=foo,uses=\{bar,baz\}]$
kohlhase-omdoc_DIA_0188	298	—	—	—	$\backslash\begin{module}[id=arith]$ $\backslash\symdef{\text{Sumfromto}}[4]{\sum_{\#1=\#2}^{\#3}{\#4}}$ $\backslash\symdef{local}{arbitraryrn}{n}$ What is the sum of the first $\\$arbitraryrn\\$ odd numbers, i.e. $\$Sumfromto{i}1arbitraryrn{2i-1}?\$$ $\backslash\end{module}$
kohlhase-omdoc_DIA_0189	301	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0190	301	—	—	—	
kohlhase-omdoc_DIA_0191	301	—	—	—	<code><axiom-inclusion xml:id=" " to=" "> </axiom-inclusion></code>
kohlhase-omdoc_DIA_0192	303	—	—	—	
kohlhase-omdoc_DIA_0193	303	—	—	—	<code>Cell [BoxData[FormBox[RowBox[{ RowBox[{ "a", " ", SuperscriptBox["x", "2"]}], "+", " ", RowBox[{ "b", " ", "x" }], " ", "+", " ", "c" }], TraditionalForm]]]</code>
kohlhase-omdoc_DIA_0194	304	—	—	—	<code>XMLElement["math", {"xmlns" -> "http://www.w3.org/1998/Math/MathML"}, {XMLElement["apply", {}, {XMLElement["plus", {}, {}], XMLElement["apply", {}, {XMLElement["times", {}, {}], XMLElement["ci", {}, {"a"}], XMLElement["apply", {}, {XMLElement["power", {}, {}], XMLElement["ci", {}, {"x"}], XMLElement["cn", {"type" -> "integer"}, {"2"}]}]}], ... }]]]</code>
kohlhase-omdoc_DIA_0195	304	—	—	—	<code>XMLElement["OMOBJ", {}, {XMLElement["OMA", {}, {XMLElement["OMS", {"cd" -> "arith1", "name" -> "plus"}, {}], XMLElement["OMA", {}, {XMLElement["OMS", {"cd" -> "arith1", "name" -> "times"}, {}], XMLElement["OMV", {"name" -> "a"}, {}], XMLElement["OMA", {}, {XMLElement["OMS", {"cd" -> "arith1", "name" -> "power"}, {}], ...}}]}]}]]]</code>
kohlhase-omdoc_DIA_0196	307	—	—	—	

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0197	308	—	—	—	
kohlhase-omdoc_DIA_0198	314	—	—	—	
kohlhase-omdoc_DIA_0199	316	—	—	—	<pre> structure list [@value] <= 0, [infixr ,100] :: (hd : @value, tl : list [@value]) </pre>
kohlhase-omdoc_DIA_0200	316	—	—	—	<pre> <adt xml:id="vf7b9f3e59-e78e-4221-8064-7fa0c5689f5d.adt" parameters="value"> <sortdef name="vf7b9f3e59-e78e-4221-8064-7fa0c5689f5d" type="free"> <constructor name="vf8a6673ac-c1d9-4698-b6ee-90213539a984" /> <constructor name="vf38164505-4983-417f-8bdc-6a42b046e933"> <argument> <type system="simpletypes"> <OMOBJ xmlns="http://www.openmath.org/OpenMath"> <OMV name="value" /> </OMOBJ> </type> <selector name="vf9fc4c672-207f-45c0-ae61-1f675fde7aed" total="yes" /> </argument> <argument> <type system="simpletypes"> <OMOBJ xmlns="http://www.openmath.org/OpenMath"> <OMA> <OMS cd="VeriFun" name="vf7b9f3e59-e78e-4221-8064-7fa0c5689f5d" /> <OMV name="value" /> </OMA> </OMOBJ> </type> <selector name="vf55767f3a-b019-4308-88f9-d68ee0db595e" total="yes" /> </argument> </constructor> </sortdef> </adt> </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0201	317	—	—	—	<pre> <presentation for=" #vf7b9f3e59-e78e-4221-8064-7fa0c5689f5d" role="applied"> <use format="VeriFun">list</use> </presentation> <presentation for=" #vf8a6673ac-c1d9-4698-b6ee-90213539a984" role="applied" bracket-style="math" precedence="1" fixity="prefix" lbrack="(" rbrack=")"> <use format="VeriFun">∅</use> </presentation> <presentation for=" #vf38164505-4983-417f-8bdc-6a42b046e933" role="applied" bracket-style="math" precedence="100" fixity="infixr" lbrack="(" rbrack=")"> <use format="VeriFun">::</use> </presentation> <presentation for=" #vf9fc4c672-207f-45c0-ae61-1f675fde7aed" role="applied" bracket-style="math" precedence="1" fixity="prefix" lbrack="(" rbrack=")"> <use format="VeriFun">hd</use> </presentation> <presentation for=" #vf55767f3a-b019-4308-88f9-d68ee0db595e" role="applied" bracket-style="math" precedence="1" fixity="prefix" lbrack="(" rbrack=")"> <use format="VeriFun">tl</use> </presentation> </pre>
kohlhase-omdoc_DIA_0202	344	—	—	—	<pre> # A RelaxNG schema for Open Mathematical documents (OMDoc 1.2: OMDoc Basic) # SYSTEM http://www.mathweb.org/omdoc/rnc/omdoc-basic.rnc # PUBLIC -//OMDoc//RNC OMDoc Basic V1.2//EN # See the documentation and examples at http://www.mathweb.org/omdoc # (c) 2004 Michael Kohlhase, released under the GNU Public License (GPL) start = omdoc include "omdocattribs.rnc" include "omdocmobj.rnc" include "omdocdoc.rnc" include "omdocdc.rnc" include "omdoecc.rnc" include "omdocmtxt.rnc" include "omdocrt.rnc" </pre>
kohlhase-omdoc_DIA_0203	345	—	—	—	<pre> # A RelaxNG for Open Mathematical documents (OMDoc 1.2: OMDoc Content Dictionaries) # SYSTEM http://www.mathweb.org/omdoc/rnc/omdoc-cd.rnc # PUBLIC -//OMDoc//RNC OMDoc CD 1.2//EN # See the documentation and examples at http://www.mathweb.org/omdoc # (c) 2004 Michael Kohlhase, released under the GNU Public License (GPL) include "omdoc-basic.rnc" include "omdocpres.rnc" include "omdocst.rnc" </pre>
kohlhase-omdoc_DIA_0204	345	—	—	—	<pre> # A RelaxNG for Open Mathematical documents (OMDoc 1.2: OMDoc Education) # SYSTEM http://www.mathweb.org/omdoc/rnc/omdoc-education.rnc # PUBLIC -//OMDoc//RNC OMDoc Education V1.2//EN # See the documentation and examples at http://www.mathweb.org/omdoc # (c) 2004 Michael Kohlhase, released under the GNU Public License (GPL) include "omdoc-mathweb.rnc" include "omdocquiz.rnc" </pre>
kohlhase-omdoc_DIA_0205	345	—	—	—	<pre> # A RelaxNG for Open Mathematical documents (OMDoc 1.2: OMDoc MathWeb) # SYSTEM http://www.mathweb.org/omdoc/rnc/omdoc-mathweb.rnc # PUBLIC -//OMDoc//RNC OMDoc MathWeb V1.2//EN # See the documentation and examples at http://www.mathweb.org/omdoc # (c) 2004 Michael Kohlhase, released under the GNU Public License (GPL) include "omdoc-cd.rnc" include "omdocext.rnc" include "omdocpf.rnc" </pre>
kohlhase-omdoc_DIA_0206	345	—	—	—	<pre> # A RelaxNG schema for Open Mathematical documents (OMDoc 1.2: OMDoc Specification) # SYSTEM http://www.mathweb.org/omdoc/rnc/omdoc-spec.rnc # PUBLIC -//OMDoc//RNC OMDoc Spec V1.2//EN # See the documentation and examples at http://www.mathweb.org/omdoc # (c) 2004 Michael Kohlhase, released under the GNU Public License (GPL) # default namespace omdoc = "http://www.mathweb.org/omdoc" include "omdoc-cd.rnc" include "omdoccth.rnc" include "omdocdg.rnc" include "omdocpf.rnc" include "omdocadt.rnc" </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0207	345	—	—	—	# A RelaxNG schema for Open Mathematical documents (OMDoc 1.2) # SYSTEM http://www.mathweb.org/omdoc/rnc/omdoc.rnc # PUBLIC -//OMDoc//RNC OMDoc V1.2//EN # See the documentation and examples at http://www.mathweb.org/omdoc # (c) 2004 Michael Kohlhase, released under the GNU Public License (GPL) include "omdoc-spec.rnc" include "omdocext.rnc" include "omdocquiz.rnc"
kohlhase-omdoc_DIA_0208	346	—	—	—	# A RelaxNG schema for Open Mathematical documents (OMDoc 1.2) Common attributes # SYSTEM http://www.mathweb.org/omdoc/rnc/omdocmobj.rnc # PUBLIC -//OMDoc//RNC OMDoc ATTRIBS V1.2//EN # See the documentation and examples at http://www.mathweb.org/omdoc # (c) 2004 Michael Kohlhase, released under the GNU Public License (GPL) default namespace omdoc = "http://www.mathweb.org/omdoc" namespace local = "" # all the explicitly namespaced attributes, except xml:lang, which # is handled explicitly nonlocal-attrs = attribute * - (local: * xml:*) {xsd:string} # the attributes for CSS and PRES styling css.attrs = attribute style {xsd:string}?, attribute class {xsd:string}? omdocref = xsd:anyURI # an URI reference pointing to an OMDoc fragment omdocrefs = list {xsd:anyURI*} # a whitespace-separated list of omdocref xref.attr = attribute xref {omdocref} idrest.attrs = css.attrs, nonlocal-attrs*, attribute xml:base {xsd:anyURI}? id.attr = attribute xml:id {xsd:ID}?, idrest.attrs omdoc.toplevel.attrs = id.attr, attribute generated-from {omdocref}? # The current XML-recommendation doesn't yet support the # three-letter short names for languages (ISO 693-2). So # the following section will be using the two-letter # (ISO 693-1) encoding for the languages. # # en : English, de : German, fr : French, # la : Latin, it : Italian, nl : Dutch, # ru : Russian, pl : Polish, es : Spanish, # tr : Turkish, zh : Chinese, ja : Japanese, # ko : Korean ... iso639 = "aa" "ab" "af" "am" "ar" "as" "ay" "az" "ba" "be" "bg" "bh" "bi" "bn" "bo" "br" "ca" "co" "cs" "cy" "da" "de" "dz" "el" "en" "eo" "es" "et" "eu" "fa" "fi" "fj" "fo" "fr" "fy" "ga" "gd" "gl" "gn" "gu" "ha" "he" "hi" "hr" "hu" "hy" "ia" "ie" "ik" "id" "is" "it" "iu" "ja" "jv" "ka" "kk" "kl" "km" "kn" "ko" "ks" "ku" "ky" "la" "ln" "lo" "lt" "lv" "mg" "mi" "mk" "ml" "mn" "mo" "mr" "ms" "mt" "my" "na" "ne" "nl" "no" "oc" "om" "or" "pa" "pl" "ps" "pt" "qu" "rm" "rn" "ro" "ru" "rw" "sa" "sd" "sg" "sh" "si" "sk" "sl" "sm" "sn" "so" "sq" "sr" "ss" "st" "su" "sv" "sw" "ta" "te" "tg" "th" "ti" "tk" "tl" "tn" "to" "tr" "ts" "tt" "tw" "ug" "uk" "ur" "uz" "vi" "vo" "wo" "xh" "yi" "yo" "za" "zh" "zu" xml.lang.attr = attribute xml:lang {iso639}?
kohlhase-omdoc_DIA_0209	347	—	—	—	# A RelaxNG schema for Open Mathematical documents (OMDoc 1.2) Module MOBJ # SYSTEM http://www.mathweb.org/omdoc/rnc/omdocmobj.rnc # PUBLIC -//OMDoc//RNC OMDoc MOBJ V1.2//EN # See the documentation and examples at http://www.mathweb.org/omdoc # (c) 2004 Michael Kohlhase, released under the GNU Public License (GPL) default namespace omdoc = "http://www.mathweb.org/omdoc" namespace om = "http://www.openmath.org/OpenMath" # we iclude the OpenMath 2 schema, but we also allow CSS attributes, etc. include "openmath2.rnc" {common.attributes = attribute id {xsd:ID}?,idrest.attrs} # we include the MathML2 schema include "mathml2/mathml2.rnc" # the legacy element, it can encapsulate the non-migrated formats legacy = (ss element legacy {id.attr, attribute formalism {xsd:anyURI}?, attribute format {xsd:anyURI}, Anything}) # to allow everything omdocmobj.class = legacy OMOBJ math

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kohlhase-omdoc_DIA_0210	347	—	—	—	<pre> # A RelaxNG schema for Open Mathematical documents (OMDoc 1.2) Module MTXT # SYSTEM http://www.mathweb.org/omdoc/dtd/omdocmtext.dtd # PUBLIC "-//OMDoc//RNC OMDoc MTXT V1.2//EN # See the documentation and examples at http://www.mathweb.org/omdoc # (c) 2004 Michael Kohlhase, released under the GNU Public License (GPL) default namespace omdoc = "http://www.mathweb.org/omdoc" omdoc.class = omtext #attribute for is a whitespace-separated list of URIsrefs fori.attrib = attribute for {omdocrefs}? from.attrib = attribute from {omdocref} verbalizes.attrib = attribute verbalizes {omdocrefs} parallel.attrib = verbalizes.attrib?,attribute index {xsd:NMTOKEN}? omdocmtxt.MC.content = metadata?,CMP* omdocmtxt.MCF.content = omdocmtxt.MC.content,FMP* # what can go into a mathematical text (to be extended in other modules) omdoc.mtext.class = text phrase term omdocmobj.class rsttype = "abstract" "introduction" "annote" "conclusion" "thesis" "comment" "antithesis" "elaboration" "motivation" "evidence" "note" "notation" statementtype = "axiom" "definition" "example" "proof" </pre>
kohlhase-omdoc_DIA_0211	348	—	—	—	<pre> "derive" "hypothesis" assertiontype = "theorem" "lemma" "corollary" "proposition" "conjecture" "false-conjecture" "obligation" "postulate" "formula" "assumption" "rule" omtext = element omtext {omdoc.toplevel.attribs, attribute type {(rsttype statementtype assertiontype)}?, attribute for {omdocref}?, attribute from {omdocref}?, verbalizes.attrib?, metadata?,CMP+,FMP*} # attribute 'for' is a URIref, to omdocdoc.class's it is needed by the 'type' attribute CMP = (ss element CMP {xml.lang.attrib, id.attrib, (omdoc.mtext.class)*}) phrase = (ss element phrase {id.attrib, parallel.attribs, attribute type {xsd:string}?, (omdoc.mtext.class)*}) # identifies a text passage and allows to attach style and role information to it term = (ss element term {id.attrib, attribute role {text}?, attribute cd {xsd:NCName}, attribute name {xsd:NCName}, (omdoc.mtext.class)*}) FMP = (ss element FMP {id.attrib, attribute logic {xsd:NMTOKEN}?, ((assumption*,conclusion*) omdocmobj.class)}) # If FMP contains a omdocmobj.class then this is the assertion, # if it contains (assumption*,conclusion*), then it is a # logical sequent (A1 ..., An - C1,...,Cm): # all the Ai entail one of the Ci assumption = (ss element assumption {id.attrib, attribute inductive {"yes" "no"}?, (omdocmobj.class)}) conclusion = (ss element conclusion {id.attrib, (omdocmobj.class?)}) </pre>
kohlhase-omdoc_DIA_0212	348	—	—	—	<pre> # A RelaxNG for Open Mathematical documents (OMDoc 1.2) Module DOC # SYSTEM http://www.mathweb.org/omdoc/rnc/omdocdoc.rnc # PUBLIC "-//OMDoc//RNC OMDoc DOC V1.2//EN # See the documentation and examples at http://www.mathweb.org/omdoc # (c) 2004 Michael Kohlhase, released under the GNU Public License (GPL) default namespace omdoc = "http://www.mathweb.org/omdoc" # extend the stuff that can go into a mathematical text omdoc.mtext.class = ignore ref ss = ignore ref omdoc.class = ss omgroup omdoc.meta.class = notAllowed metadata = element metadata {id.attrib, attribute inherits {omdocref}?, </pre>

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kohlhase-omdoc_DIA_0213	349	—	—	—	<pre> (omdoc.meta.class)* Anything = (AnyElement text)* AnyElement = element * {AnyAttribute,(text AnyElement)*} AnyAttribute = attribute * { text }* # this element can be used in lieu of a comment, it is read # by the style sheet, (comments are not) and can therefore # be transformed by them ignore = element ignore {attribute type {xsd:string}?, attribute comment {xsd:string}?, Anything} ref = element ref {id.attrib, xref.attrib, attribute type {xsd:string}??} # the types supported (there may be more over time) are # - 'include' (the default) for in-text replacement # - 'cite' for a reference with a generated label group.attrs = attribute type {xsd:anyURI}?, attribute modules {xsd:anyURI}? group.elts = metadata?,(omdoc.class)* # grouping defines the structure of a document omgroup = element omggroup {group.attrs,omdoc.toplevel.attrs,group.elts} # finally the definition of the OMDoc root element omdoc = element omdoc {omdoc.toplevel.attrs,group.attrs, attribute version {xsd:string {pattern = "1.2"}}?, group.elts} </pre>
kohlhase-omdoc_DIA_0214	349	—	—	—	<pre> # A RelaxNG schema for Open Mathematical documents (OMDoc 1.2) Module DC # SYSTEM http://www.mathweb.org/omdoc/rnc/omdocdc.rnc # PUBLIC --/OMDoc//RNC OMDoc DC V1.2//EN # See the documentation and examples at http://www.mathweb.org/omdoc # (c) 2004 Michael Kohlhase, released under the GNU Public License (GPL) default namespace dc = "http://purl.org/dc/elements/1.1/" # Persons in Dublin Core Metadata omdocdc.person.content = text # the rest of Dublin Core content omdocdc.rest.content = (text AnyElement)* omdoc.meta.class = ss dc.contributor dc.creator dc.rights dc.subject dc.title dc.description dc.publisher dc.date dc.type dc.format dc.identifier dc.source dc.language dc.relation # the MARC relator set; see http://www.loc.gov/marc/relators dcrole = attribute role {"act" "adp" "aft" "ann" "ant" "app" "aqt" "arc" "arr" "art" "asg" "asn" "att" "auc" "aud" "aui" "aus" "aut" "bdd" "bjd" "bkd" "bkp" "bnd" "bpd" "bsl" </pre>

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kohlhase-omdoc_DIA_0215	350	—	—	—	"ccp" "chr" "clb" "cli" "cll" "clt" "cmm" "cmp" "cmt" "end" "cns" "coe" "col" "com" "cos" "cot" "cov" "cpc" "cpe" "cph" "cpl" "cpt" "cre" "crp" "crr" "csl" "csp" "cst" "ctb" "cte" "ctg" "ctr" "cts" "ctt" "cur" "cwt" "dfd" "dfe" "dft" "dgg" "dis" "dln" "dnc" "dnr" "dpc" "dpt" "drm" "drt" "dsr" "dst" "dte" "dto" "dub" "edt" "egr" "elt" "eng" "etr" "exp" "fac" "flm" "fmo" "fnd" "fpv" "frg" "hnr" "hst" "ill" "ilu" "ins" "inv" "itr" "ive" "ivr" "lbt" "lee" "lel" "len" "let" "lie" "lil" "lit" "lsa" "lse" "lso" "ltg" "lyr" "mdc" "mod" "mon" "mrk" "mte" "mus" "nrt" "opn" "org" "orm" "oth" "own" "pat" "pbd" "pbl" "pfr" "pht" "plt" "pop" "ppm" "prc" "prd" "prf" "prg" "prm" "pro" "prt" "pta" "pte" "ptf" "pth" "ptt" "rbr" "rce" "rcp" "red" "ren" "res" "rev" "rpt" "rpy" "rse" "rsp" "rst" "rth" "rtm" "sad" "sce" "scl" "scr" "sec" "sgn" "sng" "spk" "spn" "spy" "srv" "stl" "stn" "str" "ths" "trc" "trl" "tyd" "tyg" "voc" "wam" "wdc" "wde" "wit" }? dclang = id.attrib , xml.lang.attrib # first the Dublin Core Metadata model of the # Dublin Metadata initiative (http://purl.org/dc) dc.contributor = element contributor {dclang,dcrole,omdocdc.person.content} dc.creator = element creator {dclang,dcrole,omdocdc.person.content} dc.title = element title {dclang,(omdoc.mtext.class)*} dc.subject = element subject {dclang,(omdoc.mtext.class)*} dc.description = element description {dclang,(omdoc.mtext.class)*} dc.publisher = element publisher {id.attrib,omdocdc.rest.content} dc.type = element type {"Dataset" "Text" "Collection"} dc.format = element format {"application/omdoc+xml"} dc.source = element source {omdocdc.rest.content} dc.language = element language {omdocdc.rest.content} dc.relation = element relation {omdocdc.rest.content} dc.rights = element rights {omdocdc.rest.content} dc.date = element date {attribute action {xsd:NMTOKEN}?, attribute who {omdocref}?, xsd:dateTime} dc.identifier = element identifier {attribute scheme {xsd:NMTOKEN},text}
kohlhase-omdoc_DIA_0216	350	—	—	—	# A RelaxNG schema for Open Mathematical documents (OMDoc 1.2) Module ST # SYSTEM http://www.mathweb.org/omdoc/rnc/omdocst.rnc # PUBLIC -//OMDoc//RNC OMDoc ST V1.2//EN # See the documentation and examples at http://www.mathweb.org/omdoc # (c) 2004 Michael Kohlhase, released under the GNU Public License (GPL) default namespace omdoc = "http://www.mathweb.org/omdoc" omdocst.scope.attrib = attribute scope {"global" "local"}? omdocst.constitutive.class = symbol axiom definition imports omdocst.nonconstitutive.class = assertion type alternative example theory

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kohlhase-omdoc_DIA_9217	351	—	—	—	<pre>theory-unique = xsd:NCName just-by.attrib = attribute just-by {omdocref} omdoc.class = omdocst.nonconstitutive.class omdocst.constitutive.attrs = id.attrib, attribute generated-from {omdocref}? sym.role.attrib = attribute role {"type" "sort" "object" "binder" "attribution" "semantic-attribution" "error"} symbol = element symbol {omdocst.scope.attrib, attribute name {theory-unique}?, omdocst.constitutive.attrs, sym.role.attrib?, metadata?,type*} axiom = element axiom {omdocst.constitutive.attrs, fori.attrib, attribute type {xsd:string}?,omdocmxt.MCF.content} #informal definitions def.informal = attribute type {"informal"}? #simple definitions exists.attrib = attribute existence {omdocref} def.simple = (attribute type {"simple"},exists.attrib?,(omdocmobj.class)) #implicit definitions unique.attrib = attribute uniqueness {omdocref} def.implicit = (attribute type {"implicit"}, exists.attrib?, unique.attrib?, FMP*) #definitions by (recursive) equations exhaust.attrib = attribute exhaustivity {omdocref} consist.attrib = attribute consistency {omdocref} def.eq = attribute type {"pattern" "inductive"}?,exhaust.attrib?,consist.attrib?, reequation+,measure?,ordering? #all definition forms, add more by extending this. defs.all = def.informal def.simple def.implicit def.eq # Definitions contain CMPs, FMPs and concept specifications. # The latter define the set of concepts defined in this element. # They can be reached under this name in the content dictionary # of the name specified in the theory attribute of the definition. definition = element definition {omdocst.constitutive.attrs, for.attrib, omdocmxt.MC.content,(defs.all)} reequation = (ss element reequation {id.attrib,omdocmobj.class,omdocmobj.class}) measure = (ss element measure {id.attrib,omdocmobj.class}) ordering = (ss element ordering {id.attrib,attribute terminating {omdocref}?,omdocmobj.class}) # the non-constitutive statements, they need a theory attribute omdoc.toplevel.attrs &= attribute theory {omdocref}? ded.status.class = " satisfiable " " counter-satisfiable " " no-consequence " " theorem " " conter-theorem " " contradictory-axioms " " tautologous-conclusion " " tautology " " equivalent " " conunter-equivalent " " unsatisfiable-conclusion " " unsatisfiable " assertion = element assertion {omdoc.toplevel.attrs, attribute type {assertiontype}?, attribute status {ded.status.class}?, attribute just-by {omdocrefs}?, omdocmxt.MCF.content} # the assertiontype has no formal meaning yet, it is solely for human consumption. # 'just-by' is a list of URIRefs that point to proof objects, etc that justifies the status.</pre>

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kohlhase-omdoc_DIA_0218	352	—	—	—	<pre> type = element type {omdoc.toplevel.attribs, just-by.attrib?, attribute system {omdocref}?, omdocmtxt.MC.content, (omdocmobj.class), (omdocmobj.class)?} alternative = element alternative {omdoc.toplevel.attribs, for .attrib, omdocmtxt.MC.content,(defs.all), ((attribute equivalence {omdocref}, attribute equivalence-thm {omdocref}) (attribute entailed-by {omdocref}, attribute entails {omdocref}, attribute entailed-by-thm {omdocref}, attribute entails-thm {omdocref})))} # just-by, points to the theorem justifying well-definedness # entailed-by, entails, point to other (equivalent definitions # entailed-by-thm, entails-thm point to the theorems justifying # the entailment relation) example = element example {omdoc.toplevel.attribs, for.attrib, attribute type {"for" "against" }?, attribute assertion {omdocref}?, omdocmtxt.MC.content, (omdocmobj.class)*} theory = element theory {id.attrib, attribute cdurl {xsd:anyURI}?, attribute cdbase {xsd:anyURI}?, attribute cdreviewdate {xsd:date}?, attribute cdversion {xsd:nonNegativeInteger}?, attribute cdrevision {xsd:nonNegativeInteger}?, attribute cdstatus {" official " " experimental" " private " " obsolete" }?, metadata?, (omdoc.class omdocst.constitutive.class tgroup)*} omdocsth.imports.model = id.attrib,from.attrib,metadata? imports = (ss element imports {omdocsth.imports.model}) tgroup = element tgroup {omdocst.constitutive.attribs.group.attribs, metadata?, (omdoc.class omdocst.constitutive.class tgroup)*} </pre>
kohlhase-omdoc_DIA_0219	352	—	—	—	<pre> # A RelaxNG schema for Open Mathematical documents (OMDoc 1.2) Module ADT # SYSTEM http://www.mathweb.org/omdoc/rnc/omdocadt.rnc # PUBLIC -//OMDoc//RNC OMDoc ADT V1.2//EN # See the documentation and examples at http://www.mathweb.org/omdoc # (c) 2004 Michael Kohlhase, released under the GNU Public License (GPL) default namespace omdoc = "http://www.mathweb.org/omdoc" omdoc.class = adt omdocadt.sym.attrib = id.attrib,omdocst.scope.attrib,attribute name {xsd:NCName} # adts are abstract data types, they are short forms for groups of symbols # and their definitions , therefore , they have much the same attributes. </pre>
kohlhase-omdoc_DIA_0220	353	—	—	—	<pre> adt = element adt {omdoc.toplevel.attribs, attribute parameters {list {xsd:NCName*}}?, metadata?, sortdef+} adttype = "loose" "generated" "free" sortdef = (ss element sortdef {omdocadt.sym.attrib, attribute role {"sort"}?, attribute type {adttype}?, metadata?,(constructor insert)*,recognizer?}) insert = (ss element insert {attribute for {omdocref}}) # for is a reference to a sort symbol element constructor = (ss element constructor {omdocadt.sym.attrib, sym.role.attrib?, metadata?,argument*}) recognizer = (ss element recognizer {omdocadt.sym.attrib, sym.role.attrib?, metadata?}) argument = (ss element argument {type,selector?}) # sort is a reference to a sort symbol element p selector = (ss element selector {omdocadt.sym.attrib, sym.role.attrib?, attribute total {"yes" "no"}?, metadata?}) </pre>

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kohlhase-omdoc_DIA_0221	353	—	—	—	<pre> # A RelaxNG schema for Open Mathematical documents (OMDoc 1.2) Module PF # SYSTEM http://www.mathweb.org/omdoc/rnc/omdocpf.rnc # PUBLIC -//OMDoc//RNC OMDoc PF V1.2//EN # See the documentation and examples at http://www.mathweb.org/omdoc # (c) 2004 Michael Kohlhase, released under the GNU Public License (GPL) default namespace omdoc = "http://www.mathweb.org/omdoc" omdocpf.opt.content = proof proofobject omdoc.class = proof proofobject proof = element proof {omdoc.toplevel.attrs, fori . attrib , metadata?,(omtext symbol definition derive hypothesis)*} proofobject = element proofobject {omdoc.toplevel.attrs, fori . attrib , metadata?,(omdocmobj.class)} omdocpf.just.content = method?, premise*, (proof proofobject)* derive.type.attr = attribute type {"conclusion" "gap"} derive = (ss element derive {id . attrib ,derive.type.attr ?, omdocmtxt.MCF.content,method?}) hypothesis = (ss element hypothesis {id . attrib , attribute inductive {"yes" "no"}?, omdocmtxt.MCF.content}) method = (ss element method {xref.attrib?, (omdocmobj.class premise proof proofobject)*}) # 'xref' is a pointer to the element defining the method </pre>
kohlhase-omdoc_DIA_0222	354	—	—	—	<pre> premise = (ss element premise {xref.attrib , attribute rank {xsd:string {pattern = "0 [1-9][0-9*"]}}?}) </pre>
kohlhase-omdoc_DIA_0223	354	—	—	—	<pre> # A RelaxNG schema for Open Mathematical documents (OMDoc 1.2) Module CTH # SYSTEM http://www.mathweb.org/omdoc/rnc/omdoccth.rnc # PUBLIC -//OMDoc//RNC OMDoc CTH V1.2//EN # See the documentation and examples at http://www.mathweb.org/omdoc # (c) 2004 Michael Kohlhase, released under the GNU Public License (GPL) default namespace omdoc = "http://www.mathweb.org/omdoc" omdocst.constitutive.class = inclusion omdoccth.imports.model &= morphism?, attribute type { "local" "global" }?, attribute conservativity {"conservative" "monomorphism" "definitional"}?, attribute conservativity —just {omdocref}? omdoc.toplevel.attrs &= attribute generated —via {omdocref}? omdocst.constitutive.attrs &= attribute generated —via {omdocref}? omdoc.class = theory —inclusion axiom —inclusion omdoccth.theory —inclusion.justification = obligation* omdoccth.axiom —inclusion.justification = obligation* fromto.attrib = from.attrib, attribute to {omdocref} # attributes 'to' and 'from' are URIref morphism = (ss element morphism {id.attrib,attribute hiding {omdocrefs}?, attribute base {omdocrefs}?,def.eq?}) # base points to some other morphism it extends inclusion = element inclusion {id . attrib ,attribute via {omdocref}} # via points to a theory —inclusion theory —inclusion = element theory —inclusion {omdoc.toplevel.attrs,fromto.attrib, metadata?,morphism?, (omdoccth.theory —inclusion.justification)} axiom —inclusion = element axiom —inclusion {omdoc.toplevel.attrs,fromto.attrib, metadata?,morphism?, (omdoccth.axiom —inclusion.justification)} obligation = (ss element obligation {id . attrib , attribute induced —by {omdocref}, attribute assertion {omdocref}}) # attribute 'assertion' is a URIref, points to an assertion # that is the proof obligation induced by the axiom or definition # specified by 'induced —by'. </pre>

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kohlhase-omdoc_DIA_0224	355	—	—	—	# A RelaxNG schema for Open Mathematical documents (OMDoc 1.2) Module DOC # SYSTEM http://www.mathweb.org/omdoc/rnc/omdocdoc.rnc # PUBLIC -//OMDoc//RNC OMDoc RT V1.2//EN # See the documentation and examples at http://www.mathweb.org/omdoc # (c) 2004 Michael Kohlhase, released under the GNU Public License (GPL) default namespace omdoc = "http://www.mathweb.org/omdoc" omdoc.mtext.class = ss ul ol dl p note link table idx omdocrt.common.attrib = id.attrib, fori.attrib, parallel.attribs ul = element ul {omdocrt.common.attrib, metadata?,li+} ol = element ol {omdocrt.common.attrib, metadata?,li+} dl = element dl {omdocrt.common.attrib, metadata?,di+} li = element li {omdocrt.common.attrib, metadata?,(omdoc.mtext.class)*} di = element di {omdocrt.common.attrib, metadata?,dt+,dd*} dt = element dt {omdocrt.common.attrib, metadata?,(omdoc.mtext.class)*} dd = element dd {omdocrt.common.attrib, metadata?,(omdoc.mtext.class)*} p = element p {omdocrt.common.attrib, (omdoc.mtext.class)*} note = element note {omdocrt.common.attrib, attribute type {xsd:NMTOKEN}?, (omdoc.mtext.class)*} # a simplified html table table = element table {omdocrt.common.attrib, tr+} tr = (ss element tr {omdocrt.common.attrib, (td th)+}) td = (ss element td {omdocrt.common.attrib, (omdoc.mtext.class)*}) th = (ss element th {omdocrt.common.attrib, (omdoc.mtext.class)*}) link = element link {omdocrt.common.attrib, attribute href {xsd:anyURI}, (omdoc.mtext.class)*} # index index.att = attribute sort-by {text}?, attribute see {omdocrefs}?, attribute seealso {omdocrefs}?, attribute links { list {xsd:anyURI*}}? idx = element idx {(id.attrib xref.attrib), idt?, ide+} ide = element ide {attribute index {xsd:NCName}, index.att, idp*} idt = element idt {idrest.attribs, omdoc.mtext.class*} idp = element idp {index.att, omdoc.mtext.class*}

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kohlhase-omdoc_DIA_0225	356	—	—	—	<pre> # A RelaxNG schema for Open Mathematical documents (OMDoc 1.2) Module EXT # SYSTEM http://www.mathweb.org/omdoc/rnc/omdocext.rnc # PUBLIC -//OMDoc//RNC OMDoc EXT V1.2//EN # See the documentation and examples at http://www.mathweb.org/omdoc # (c) 2004 Michael Kohlhase, released under the GNU Public License (GPL) default namespace omdoc = "http://www.mathweb.org/omdoc" omdoc.mtext.class = omlet omdocext.class = private code omlet omdoc.class = omdocext.class omdocext.private.attrib = fori.attrib, attribute requires {omdocref}? private = element private {omdoc.toplevel.attrs,omdocext.private.attrib, attribute reformulates {omdocref}?, metadata?,data+} # reformulates is a URIref to the omdoc elements that are reformulated by the # system-specific information in this element code = element code {omdoc.toplevel.attrs,omdocext.private.attrib, (metadata?,data+,input?,output?,effect?)} input = (ss element input {id.attrib, (omdocmtxt.MCF.content)}) output = (ss element output {id.attrib, (omdocmtxt.MCF.content)}) effect = (ss element effect {id.attrib, (omdocmtxt.MCF.content)}) data = (ss element data {id.attrib, attribute format {xsd:string}?, attribute href {xsd:anyURI}?, attribute size {xsd:string}?, attribute pto {xsd:string}?, attribute pto-version {xsd:string}?, attribute original {"external" "local"}?, Anything}) omlet = (ss element omlet {id.attrib, attribute action {"display" "execute" "other"}?, attribute show {"new" "replace" "embed" "other"}?, attribute actuate {"onPresent" "onLoad" "onRequest" "other"}?, (omdoc.mtext.class param)*, (attribute data {xsd:anyURI} (private code))) param = (ss element param {id.attrib, attribute name {xsd:string}, attribute value {xsd:string}?, attribute valuetype {"data" "ref" "object"}?, (omdocmobj.class)?}) </pre>
kohlhase-omdoc_DIA_0226	356	—	—	—	<pre> # A RelaxNG for Open Mathematical documents (OMDoc 1.2) Module PRES # SYSTEM http://www.mathweb.org/omdoc/rnc/omdocpres.rnc # PUBLIC -//OMDoc//RNC OMDoc PRES 1.2//EN # See the documentation and examples at http://www.mathweb.org/omdoc # (c) 2004 Michael Kohlhase, released under the GNU Public License (GPL) default namespace omdoc = "http://www.mathweb.org/omdoc" </pre>

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kohlhase-omdoc_DIA_0227	357	—	—	—	<pre> # we include the OpenMath 2 schema, but we also allow CSS attributes, etc. include "xslt10.rnc" #xslt10 = external "xslt10.rnc" omdoc.class = presentation omstyle crossref.attrib = attribute crossref-symbol {"no" "yes" "brackets" "separator" "lbrack" "rbrack" "all"} fixity.attrib = attribute fixity {"prefix" "infix" "postfix" "assoc" "infixl" "infixr"} format.attrib = attribute format {xsd:string}, attribute requires {omdocref}?, xml.lang.attrib bracket-style.attrib = attribute bracket-style {"lisp" "math"} lbrack.attrib = attribute lbrack {xsd:string} rbrack.attrib = attribute rbrack {xsd:string} separator.attrib = attribute separator {xsd:string} precedence.attrib = attribute precedence {xsd:string {pattern = "0 [1-9][0-9]*"}} role.attrib = attribute role {"applied" "binding" "key"} presentation = element presentation {omdoc.toplevel.attrs, attribute for {omdocref}, role.attrib?, (xref.attrib? (fixity.attrib?, lbrack.attrib?, rbrack.attrib?, separator.attrib?, bracket-style.attrib?, precedence.attrib?, crossref.attrib?, CMP*, (use xslt style)*))} omdocpres.use.mix = elt txt recurse value-of map use = (ss element use {format.attrib, bracket-style.attrib?, fixity.attrib?, lbrack.attrib?, rbrack.attrib?, separator.attrib?, attribute element {xsd:string}?, attribute attributes {xsd:string}?, crossref.attrib?, (text omdocpres.use.mix)*}) # the attributes in the <use> element overwrite those in the # <presentation> element, therefore, they do not have defaults omstyle = element omstyle {omdoc.toplevel.attrs, attribute for {omdocref}?, attribute element {xsd:string}?, (xslt style)*} xslt = (ss element xslt {format.attrib, xref.attrib?, template.model}) style = (ss element style {format.attrib, (omdocpres.use.mix)*}) # this element contains mock xslt expressed in the elements below elt = (ss element element {attribute name {xsd:NMTOKEN}, attribute crid {xsd:string}?, attribute cr {"yes" "no"}?, attribute ns {xsd:anyURI}?, (attrb omdocpres.use.mix)*}) map = (ss element map {attribute select {xsd:string}?, attribute lbrack {xsd:string}?, attribute rbrack {xsd:string}?, attribute precedence {xsd:string}?, </pre>
kohlhase-omdoc_DIA_0228	358	—	—	—	<pre> separator?, (omdocpres.use.mix)*}) separator = (ss element separator {(omdocpres.use.mix)*}) attrb = (ss element attribute {attribute name {xsd:NMTOKEN}, attribute ns {xsd:anyURI}?, (attribute select {xsd:string} (txt value-of)*)) txt = (ss element text {text}) value-of = (ss element value-of {attribute select {xsd:string}}) recurse = (ss element recurse {attribute select {xsd:string}??}) </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0229	358	—	—	—	<pre> # A RelaxNG schema for Open Mathematical documents (OMDoc 1.2) Module QUIZ # SYSTEM http://www.mathweb.org/omdoc/rnc/omdocquiz.rnc # PUBLIC -//OMDoc//RNC OMDoc QUIZ V1.2//EN # See the documentation and examples at http://www.mathweb.org/omdoc # (c) 2004 Michael Kohlhase, released under the GNU Public License (GPL) default namespace omdoc = "http://www.mathweb.org/omdoc" omdoc.class = exercise hint mc solution exercise = element exercise {id.attrib, fori.attrib, omdocmtxt.MCF.content, hint*, (solution* mc*)} omdocpf.opt.content = notAllowed hint = element hint {omdoc.toplevel.attrs, fori.attrib, omdocmtxt.MCF.content} solution = element solution {omdoc.toplevel.attrs, fori.attrib, metadata?, (omdoc.class)*} mc = element mc {omdoc.toplevel.attrs, fori.attrib, choice, hint?, answer} choice = (ss element choice {id.attrib, omdocmtxt.MCF.content}) answer = (ss element answer {id.attrib, attribute verdict {"true" "false"}?, omdocmtxt.MCF.content}) </pre>
kohlhase-omdoc_DIA_0230	359	—	—	—	<pre> # RELAX NG Schema for OpenMath 2 default namespace om = "http://www.openmath.org/OpenMath" #start = OMOBJ # OpenMath object constructor OMOBJ = element OMOBJ { compound.attributes, attribute version { xsd:string }?, omel } # Elements which can appear inside an OpenMath object omel = OMS OMV OMI OMB OMSTR OMF OMA OMBIND OME OMATTR OMR # things which can be variables omvar = OMV attvar attvar = element OMATTR { common.attributes, (OMATP, (OMV attvar))} cbase = attribute cbase { xsd:anyURI}? # attributes common to all elements common.attributes = (attribute id { xsd:ID })? # attributes common to all elements that construct compound OM objects. compound.attributes = common.attributes, cbase # symbol OMS = element OMS { common.attributes, attribute name {xsd:NCName}, attribute cd {xsd:NCName}, </pre>

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kohlhase-omdoc_DIA_0232	361	—	—	—	<pre> # A RelaxNG schema for MathML2 # SYSTEM http://www.mathweb.org/omdoc/rnc/mathml2.rnc # (c) 2005 Michael Kohlhase, released under the GNU Public License (GPL) default namespace m = "http://www.w3.org/1998/Math/MathML" namespace a = "http://relaxng.org/ns/compatibility/annotations/1.0" namespace xlink = "http://www.w3.org/1999/xlink" namespace local = "" non-mathml-attrs = attribute * - (local:* xlink:href) {xsd:string} MathML.Common.attr = attribute class {xsd:NMTOKENS}?, attribute style {xsd:string}?, attribute id {xsd:ID}?, attribute xlink:href {xsd:anyURI}?, non-mathml-attrs* include "mathml2-presentation.rnc" include "mathml2-content.rnc" Presentation-expr.class = PresExpr.class ContExpr.class Content-expr.class = ContExpr.class PresExpr.class PresExpr.class = Presentation-token.class Presentation-layout.class Presentation-script.class Presentation-table.class mspace maction merror mstyle ContExpr.class = Content-tokens.class Content-arith.class Content-functions.class Content-logic.class Content-constants.class Content-sets.class Content-relations.class Content-elementary-functions.class Content-calculus.class Content-linear-algebra.class Content-vector-calculus.class Content-statistics.class Content-constructs.class semantics Browser-interface.attr = attribute baseline {xsd:string}?, [a:default = "scroll"] attribute overflow {"scroll" "elide" "truncate" "scale"}?, attribute altimg {xsd:anyURI}?, attribute alttext {xsd:string}?, attribute type {xsd:string}?, attribute name {xsd:string}?, attribute height {xsd:string}?, attribute width {xsd:string}? math.attlist = Browser-interface.attr, attribute macros {xsd:string}?, [a:default = "inline"] </pre>
kohlhase-omdoc_DIA_0233	362	—	—	—	<pre> attribute display {"block" "inline"}?, MathML.Common.attr math.content = PresExpr.class ContExpr.class math = element math {math.attlist,math.content*} </pre>
kohlhase-omdoc_DIA_0234	362	—	—	—	<pre> math.content = PresExpr.class ContExpr.class math = element math {math.attlist,math.content*} </pre>

Identifier	Page	Conf.	LaTeX source	Rendered	Image
kohlhase-omdoc_DIA_0235	362	—	—	—	# A RelaxNG schema for MathML2 Presentation Elements # SYSTEM http://www.mathweb.org/omdoc/rnc/mathml2-content.rnc # (c) 2005 Michael Kohlhase, released under the GNU Public License (GPL) default namespace m = "http://www.w3.org/1998/Math/MathML" namespace a = "http://relaxng.org/ns/compatibility/annotations/1.0" # Simple sizes simple-size= "small" "normal" "big" # Centering values centering.values = "left" "center" "right" # The named spaces # this is also used in the value of the "width" attribute on the "mpadded" element named-space = "veryverythinmathspace" "verythinmathspace" "thinmathspace" "mediummathspace" "thickmathspace" "verythickmathspace" "veryverythickmathspace" # Thickness thickness = "thin" "medium" "thick" # number with units used to specified lengths length-with-unit = xsd:string # {pattern="(-(?[0-9]+ [0-9]*\.[0-9]+)*(em ex px in cm mm pt pc %) 0)"} length-with-optional-unit = xsd:string # {pattern="(-(?[0-9]+ [0-9]*\.[0-9]+)*(em ex px in cm mm pt pc %)?)?"} # This is just "infinity" that can be used as a length infinity = "infinity" # colors defined as RGB RGB-color = xsd:string {pattern="#((([0-9] [a-f]){3} ([0-9] [a-f]){6}))"} # This schema module defines sets of attributes common to several elements # of presentation MathML. # The mathematics style attributes. These attributes are valid on all # presentation token elements except "mspace" and "mglyph", and on no # other elements except "mstyle". Token-style.attrib = attribute mathvariant { "normal" "bold" "italic" "bold-italic" "double-struck" "bold-fraktur" "script" "bold-script" "fraktur" "sans-serif" "bold-sans-serif" "sans-serif-italic" "sans-serif-bold-italic" "monospace" } }, attribute mathsize {simple-size length-with-unit}?, # For both of the following attributes the types should be more restricted attribute mathcolor {xsd:string}?, attribute mathbackground {xsd:string}?

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kohlhase-omdoc_DIA_0236	363	—	—	—	# These operators are all related to operators. They are valid on "mo" and "mstyle". Operator.attrib = # this attribute value is normally inferred from the position of # the operator in its "<mrow"> attribute form {"prefix" "infix" "postfix"}?, # set by dictionary, else it is "thickmathspace" attribute lspace {length-with-unit named-space}?, # set by dictionary, else it is "thickmathspace" attribute rspace {length-with-unit named-space}?, # set by dictionary, else it is "false" attribute fence {xsd:boolean}?, # set by dictionary, else it is "false" attribute separator {xsd:boolean}?, # set by dictionary, else it is "false" attribute stretchy {xsd:boolean}?, # set by dictionary, else it is "true" attribute symmetric {xsd:boolean}?, # set by dictionary, else it is "false" attribute movablelimits {xsd:boolean}?, # set by dictionary, else it is "false" attribute accent {xsd:boolean}?, # set by dictionary, else it is "false" attribute largeop {xsd:boolean}?, attribute minsize {length-with-unit named-space}?, attribute maxsize {length-with-unit named-space infinity xsd:float}? mglyph = element mglyph {attribute alt {xsd:string}?, attribute fontfamily {xsd:string}?, attribute index {xsd:positiveInteger }?} # This is the XML schema module for the token elements of the # presentation part of MathML. Glyph-alignmark.class = malignmark mglyph # "mi" is supposed to have a default value of its "mathvariant" attribute set to "italic " mi = element mi {Token-style.attrib, MathML.Common.attrib,(Glyph-alignmark.class text)*} # "mo" mo = element mo {Operator.attrib,Token-style.attrib,MathML.Common.attrib, (text Glyph-alignmark.class)*} # "mn" mn = element mn {Token-style.attrib, MathML.Common.attrib,(text Glyph-alignmark.class)*} # "mtext" mtext = element mn {Token-style.attrib, MathML.Common.attrib,(text Glyph-alignmark.class)*} # ms (the values of "lquote" or "rquote" are not restricted to be one character strings ...) ms = element ms {[a:default="&quot;"] attribute lquote {xsd:string}?, [a:default="&quot;"] attribute rquote {xsd:string}?, Token-style.attrib,MathML.Common.attrib, (text Glyph-alignmark.class)*} # And the group of any token Presentation-token.class = mi mo mn mtext ms # This is an XML Schema module for the presentation elements of MathML # dealing with subscripts and superscripts. # "msub" msub = element msub {attribute subscriptshift {length-with-unit}?, MathML.Common.attrib, Presentation-expr.class,(Presentation-expr.class)} # "msup" msup = element msup {attribute superscriptshift {length-with-unit}?, MathML.Common.attrib,

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kohlhase-omdoc_DIA_0237	364	—	—	—	Presentation—expr.class,Presentation—expr.class} # "msubsup" msubsup = element msubsup {MathML.Common.attrib, attribute subscriptshift {length—with—unit}?, attribute superscriptshift {length—with—unit}?, Presentation—expr.class,Presentation—expr.class} # "munder" munder = element munder {MathML.Common.attrib, attribute accentunder {xsd:boolean}?, Presentation—expr.class,Presentation—expr.class} # "mover" mover = element mover {MathML.Common.attrib, attribute accent {xsd:boolean}?, Presentation—expr.class,Presentation—expr.class} # "munderover" munderover = element munderover {MathML.Common.attrib, attribute accentunder {xsd:boolean}?, attribute accent {xsd:boolean}?, Presentation—expr.class, Presentation—expr.class} # "mmultiscripts", "mprescripts" and "none" Presentation—expr—or—none.class = Presentation—expr.class none mmultiscripts = element mmultiscripts{MathML.Common.attrib, Presentation—expr.class, (Presentation—expr—or—none.class, Presentation—expr—or—none.class)*, (mprescripts, (Presentation—expr—or—none.class, Presentation—expr—or—none.class)*)?} none = element none {empty} mprescripts = element mprescripts {empty} Presentation—script.class = msub msup msubsup munder mover munderover mmultiscripts mspace = element mspace {[a:defaultValue = "0em"] attribute width {length—with—unit named—space}?, [a:defaultValue = "0ex"] attribute height {length—with—unit}?, [a:defaultValue = "0ex"] attribute depth {length—with—unit}?, [a:defaultValue="auto"] attribute linebreak {"auto" "newline" "indentingnewline" "nobreak" "goodbreak" "badbreak"}?, MathML.Common.attrib} # This is the XML schema module for the layout elements of the # presentation part of MathML. # "mrow" mrow = element mrow {MathML.Common.attrib,(Presentation—expr.class)*} # "mfrac" mfrac = element mfrac {attribute bevelled {xsd:boolean}?, [a:defaultValue = "center"] attribute denomalign {centering.values}?, [a:defaultValue = "center"] attribute numalign {centering.values}?, [a:defaultValue="1"]

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kohlhase-omdoc_DIA_0238	365	—	—	—	<pre> attribute linethickness {length-with-optional-unit thickness}?, MathML.Common.attrib, Presentation-expr.class,Presentation-expr.class} # "msqrt" msqrt = element msqrt {MathML.Common.attrib,(Presentation-expr.class)*} # "mroot" mroot = element mroot {MathML.Common.attrib,Presentation-expr.class,Presentation-expr.class} # "mpadded" mpadded-space = xsd:string {pattern="(\+ -)?([0-9]+ [0-9]*\.[0-9]+)((%?)*((width space height depth)))(em ex px in cm mm pt pc))"} # MaxF: definition from spec seems wrong, # fixing to ([+ -] unsigned-number (%[pseudo-unit]) pseudo-unit h-unit)) namespace 0 mpadded-width-space = xsd:string {pattern="((\+ -)?([0-9]+ [0-9]*\.[0-9]+)((%?)*((width space height depth)?)) (width space height depth))(em ex px in cm mm pt pc))} ((veryverythin verythin thin medium thick verythick veryverythick mathspace) 0"} mpadded = element mpadded {attribute width {mpadded-width-space}, # should have default=0 below but '0' is not in value space # see bug #425 attribute lspace {mpadded-space}?, attribute height {mpadded-space}?, attribute depth {mpadded-space}?, MathML.Common.attrib, (Presentation-expr.class)*} # "mphantom" mphantom = element mphantom.attlist {MathML.Common.attrib,Presentation-expr.class*} # "mfenced" mfenced = element mfenced {[a:defaultValue= "("] attribute open {xsd:string}?, [a:defaultValue = ")"] attribute close {xsd:string}?, [a:defaultValue = ","] attribute separators {xsd:string}?, MathML.Common.attrib, (Presentation-expr.class)*} # "menclose" menclose = element menclose {[a:defaultValue="longdiv"] attribute notation {"actuarial" "longdiv" "radical" "box" "roundedbox" "circle" "left" "right" "top" "bottom" "updiagonalstrike" "downdiagonalstrike" "verticalstrike" "horizontalstrike" }?, MathML.Common.attrib, (Presentation-expr.class)*} # And the group of everything Presentation-layout.class = mrow mfrac msqrt mroot mpadded mphantom mfenced menclose # This is an XML Schema module for tables in MathML presentation. Table-alignment.attrib = [a:defaultValue = "baseline"] attribute rowalign {xsd:string {pattern="(top bottom center baseline axis)(top bottom center baseline axis)*"} }?, [a:defaultValue = "center"] attribute columnalign {xsd:string {pattern="(left center right)(left center right)*"} }?, attribute groupalign {xsd:string}? mtr = element mtr {Table-alignment.attrib, MathML.Common.attrib,mtd+} mlabeledtr = element mlabeledtr {Table-alignment.attrib,MathML.Common.attrib,mtd*} # "mtd" </pre>

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kohlhase-omdoc_DIA_0239	366	—	—	—	mtd = element mtd {Table−alignment.attrib, [a:defaultValue="1"] attribute colspan {xsd:positiveInteger}?, [a:defaultValue="1"] attribute rowspan {xsd:positiveInteger}?, MathML.Common.attrib, Presentation−expr.class*} # "mtable" mtable = element mtable {Table−alignment.attrib, [a:defaultValue="axis"] attribute align {xsd:string}?, [a:defaultValue="true"] attribute alignmentscope {xsd:string {pattern="(true false)(true false)*"}?}, [a:defaultValue="auto"] attribute columnwidth {xsd:string}?, [a:defaultValue="auto"] attribute width {xsd:string}?, [a:defaultValue="1.0ex"] attribute rowspacing {xsd:string}?, [a:defaultValue="0.8em"] attribute columnspacing {xsd:string}?, [a:defaultValue="none"] attribute rowlines {xsd:string}?, [a:defaultValue="none"] attribute columlines {xsd:string}?, [a:defaultValue="none"] attribute frame {"none" "solid" "dashed"}?, [a:defaultValue="0.4em 0.5ex"] attribute framespacing {xsd:string}?, [a:defaultValue="false"] attribute equalrows {xsd:boolean}?, [a:defaultValue="false"] attribute equalcolumns {xsd:boolean}?, [a:defaultValue="false"] attribute displaystyle {xsd:boolean}?, [a:defaultValue="right"] attribute side {" left " "right " " leftoverlap " " rightoverlap " }?, [a:defaultValue="0.8em"] attribute minlabelspacing {length−with−unit}?, MathML.Common.attrib, (mtr mabeledtr)*} # "maligngroup" maligngroup = element maligngroup { attribute groupalign {" left " " center " " right " " decimalpoint" }?, MathML.Common.attrib} # "malignmark" malignmark = element malignmark {[a:defaultValue="left"] attribute edge {"left" "right"}?, MathML.Common.attrib} Presentation−table.class = mtable maligngroup malignmark # "mstyle" mstyle = element mstyle {attribute scriptlevel {xsd:integer}?, attribute displaystyle {xsd:boolean}?, [a:defaultValue="0.71"] attribute scriptsizemultiplier {xsd:decimal}?, [a:defaultValue="8pt"] attribute scriptminsize {length−with−unit}?, attribute color {xsd:string}?, [a:defaultValue="transparent"] attribute background {xsd:string}?, [a:defaultValue="0.0555556em"] attribute veryverythinmathspace {length−with−unit}?, [a:defaultValue="0.111111em"] attribute verythinmathspace {length−with−unit}?, [a:defaultValue="0.166667em"] attribute thinmathspace {length−with−unit}?, [a:defaultValue="0.222222em"] attribute mediummathspace {length−with−unit}?, [a:defaultValue="0.277778em"] attribute thickmathspace {length−with−unit}?, [a:defaultValue="0.333333em"] attribute verythickmathspace {length−with−unit}?, [a:defaultValue="0.388889em"] attribute veryverythickmathspace {length−with−unit}?, [a:defaultValue="1"] attribute linethickness {length−with−optional−unit thickness}?, Operator.attrib,Token−style.attrib,MathML.Common.attrib, Presentation−expr.class*} # This is the XML Schema module for the MathML "merror" element. merror = element merror {MathML.Common.attrib,Presentation−expr.class*} # This is the XML Schema module for the MathML "maction" element. maction = element maction {attribute actiontype {xsd:string}?, [a:defaultValue="1"] attribute selection {xsd:positiveInteger}?, MathML.Common.attrib, Presentation−expr.class*}

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kohlhase-omdoc_DIA_0240	367	—	—	—	# A RelaxNG schema for MathML2 Content Elements # SYSTEM http://www.mathweb.org/omdoc/rnc/mathml2-content.rnc # (c) 2005 Michael Kohlhase, released under the GNU Public License (GPL) default namespace m = "http://www.w3.org/1998/Math/MathML" namespace a = "http://relaxng.org/ns/compatibility/annotations/1.0" Definition .attrib = attribute encoding {xsd:string}?, attribute definitionURL {xsd:anyURI}? # This is the XML schema module for the token elements of the content part of MathML. Content-token.content = text Presentation-expr.class # the content of "cn" may have <sep> elements in it sep = element sep {empty} cn = element cn {#####attribute base {xsd:positiveInteger [1,...,36]}, attribute type {"e-notation" "integer" "rational" "real" "complex-cartesian" "complex-polar" "constant" }?, Definition .attrib , MathML.Common.attrib, (text sep Presentation-expr.class)*} # "ci" ci = element ci {attribute type {xsd:string}?, Definition .attrib , MathML.Common.attrib, Content-token.content} # "csymbol" csymbol = element csymbol {Definition.attrib, MathML.Common.attrib, Content-token.content} # And the group of everything Content-tokens.class = cn ci csymbol # This is an XML Schema module for the "arithmetic" operators of content MathML. Arith.type = Definition.attrib, MathML.Common.attrib abs = element abs {Arith.type} conjugate = element conjugate {Arith.type} arg = element arg {Arith.type} real = element real {Arith.type} imaginary = element imaginary{Arith.type} floor = element floor {Arith.type} ceiling = element floor {Arith.type} power = element power {Arith.type} root = element root {Arith.type} minus = element minus {Arith.type} plus = element plus {Arith.type} sum = element sum {Arith.type} times = element times {Arith.type} product = element product {Arith.type} max = element max {Arith.type} min = element min {Arith.type} factorial = element factorial {Arith.type} quotient = element quotient {Arith.type} divide = element divide {Arith.type} rem = element rem {Arith.type} gcd = element gcd {Arith.type} lcm = element lcm {Arith.type}

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kohlhase-omdoc_DIA_0241	368	—	—	—	Content—arith.class = abs conjugate factorial arg real imaginary floor ceiling quotient divide rem minus plus times power root max min gcd lcm sum product # This is an XML Schema module for operators dealing with functions in content MathML. Functions.type = Definition.attrib, MathML.Common.attrib # "compose" compose = element compose {Functions.type} # Domain, codomain and image domain = element domain {Functions.type} codomain = element codomain {Functions.type} image = element image {Functions.type} # "domainofapplication" domainofapplication = element domainofapplication{Definition.attrib, MathML.Common.attrib, Content—expr.class} # identity ident = element ident {Functions.type} Content—functions.class = compose domain codomain image domainofapplication ident # This is an XML Schema module for the logic operators of content MathML. Logic.type = Definition.attrib, MathML.Common.attrib and = element and {Logic.type} or = element or {Logic.type} xor = element xor {Logic.type} not = element not {Logic.type} exists = element exists {Logic.type} forall = element forall {Logic.type} implies = element implies {Logic.type} Content—logic.class = and or xor not exists forall implies # This is an XML Schema module for the basic constructs of content MathML. apply = element apply {MathML.Common.attrib, Content—expr.class*} interval = element interval {MathML.Common.attrib, [a:defaultvalue = "closed"] attribute closure {"closed" "open" "open—closed" "closed—open"}?, (Content—expr.class),(Content—expr.class)} inverse = element inverse {Definition.attrib, MathML.Common.attrib} condition = element condition {Definition.attrib, Content—expr.class+} declare = element declare {attribute type {xsd:string}?, attribute scope {xsd:string}?, attribute nargs {xsd:nonNegativeInteger}?, attribute occurrence {"prefix" "infix" "function—model"}?, Definition.attrib, Content—expr.class+} lambda = element lambda {MathML.Common.attrib, Content—expr.class+} # "piecewise" and its inner elements otherwise = element otherwise {Content—expr.class, MathML.Common.attrib} piece = element piece {MathML.Common.attrib, Content—expr.class+} piecewise = element piecewise {MathML.Common.attrib, piece*, (otherwise, piece)*} bvar = element bvar {MathML.Common.attrib, Content—expr.class+} degree = element degree {MathML.Common.attrib, Content—expr.class+} Content—constructs.class = apply interval inverse condition declare lambda piecewise bvar degree

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kohlhase-omdoc_DIA_0242	369	—	—	—	<pre> # This is the XML Schema module for the basic constants of MathML content. Constant.type = Definition.attrib, MathML.Common.attrib # Basic sets naturalnumbers= element naturalnumbers {Constant.type} primes= element primes{Constant.type} integers = element integers {Constant.type} rationals = element rationals{Constant.type} reals = element reals {Constant.type} complexes = element complexes {Constant.type} #Empty set emptyset = element emptyset {Constant.type} # Basic constants exponentiale = element exponentiale {Constant.type} imaginaryi = element imaginaryi {Constant.type} pi = element pi {Constant.type} eulergamma = element eulergamma {Constant.type} # Boolean constants true = element true {Constant.type} false = element false {Constant.type} # Infinty infini = element infinity {Constant.type} # NotANumber notanumber = element notanumber {Constant.type} Content-constants.class = naturalnumbers primes integers rationals reals complexes emptyset exponentiale imaginaryi pi eulergamma true false infini notanumber # This is an XML Schema module for the elementary functions in content MathML. Elementary-functions.type = Definition.attrib,MathML.Common.attrib # Exp and logs exp= element exp {Elementary-functions.type} ln = element ln {Elementary-functions.type} log = element log {Elementary-functions.type} # special element of the base of logarithms logbase = element logbase {MathML.Common.attrib,Content-expr.class} # Trigonometric functions sin = element sin {Elementary-functions.type} cos = element cos {Elementary-functions.type} tan = element tan {Elementary-functions.type} sec = element sec {Elementary-functions.type} csc = element csc {Elementary-functions.type} cot = element cot {Elementary-functions.type} arcsin = element arcsin {Elementary-functions.type} arccos = element arccos {Elementary-functions.type} arctan = element arctan {Elementary-functions.type} arccot = element arccot {Elementary-functions.type} arccsc = element arccsc {Elementary-functions.type} arcsec = element arcsec {Elementary-functions.type} # Hyperbolic trigonometric functions sinh = element sinh {Elementary-functions.type} </pre>

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kohlhase-omdoc_DIA_0243	370	—	—	—	cosh = element cosh {Elementary-functions.type} tanh = element tanh {Elementary-functions.type} sech = element sech {Elementary-functions.type} csch = element csch {Elementary-functions.type} coth = element coth {Elementary-functions.type} arccosh = element arccosh {Elementary-functions.type} arccoth = element arccoth {Elementary-functions.type} arccsch = element arccsch {Elementary-functions.type} arcsech = element arcsech {Elementary-functions.type} arcsinh = element arcsinh {Elementary-functions.type} arctanh = element arctanh {Elementary-functions.type} # And the group of everything Content-elementary-functions.class = exp ln log logbase sin cos tan sec csc cot arcsin arccos arctan arcsec arccsc arccot sinh cosh tanh sech csch coth arccosh arccoth arccsch arcsech arcsinh arctanh # This is an XML Schema module for the relational operators of content MathML. # a common type for all this Relations.type = Definition.attrib , MathML.Common.attrib eq = element eq {Relations.type} neq = element neq {Relations.type} leq = element leq {Relations.type} lt = element lt {Relations.type} geq = element geq {Relations.type} gt = element gt {Relations.type} equivalent = element equivalent {Relations.type} approx = element approx {Relations.type} factorof = element factorof {Relations.type} # And the group of everything Content-relations.class = eq neq leq lt geq gt equivalent approx factorof # "annotation" annotation = element annotation {attribute encoding {xsd:string}?,MathML.Common.attrib,text} # "annotation-xml" anyElement = element * {(attribute * {text}) text anyElement}*} annotation-xml = element annotation-xml {Definition.attrib,MathML.Common.attrib, anyElement} # "semantics" semantics = element semantics {attribute encoding {xsd:string}?, attribute definitionURL {xsd:anyURI}?, MathML.Common.attrib, Content-expr.class, (annotation annotation-xml)*} # This is an XML Schema module for the part of content MathML dealing with sets and lists. # "set" ("type" could be "multiset" or "normal" or anything else) set = element set {attribute type {xsd:string}?, MathML.Common.attrib, Content-expr.class*} # "list" lst = element list {attribute order {"lexicographic" "numeric"}?, MathML.Common.attrib, Content-expr.class*} # "union" union = element union {Definition.attrib, MathML.Common.attrib,Content-expr.class*} intersect = element intersect {Definition.attrib, MathML.Common.attrib,Content-expr.class*} in = element in {Definition.attrib, MathML.Common.attrib} notin = element notin {Definition.attrib, MathML.Common.attrib} subset = element subset {Definition.attrib, MathML.Common.attrib} prsubset = element prsubset {Definition.attrib, MathML.Common.attrib}

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kohlhase-omdoc_DIA_0245	387	—	—	—	m:annotation – element, 115 m:annotation-xml – element, 17, 115, 199 answer – element, 210, 321 ant – attribute value – for role on dc:*, 101 antecedent – bibliographic, 101 – scientific, 101 antithesis – attribute value – for type on omtext, 125 mq:anyorder – attribute, 255–257 applet 205 – Java, 207 application 18, 108, 115 – XML, 6 – web, 264 – XML, 3, 14, 85, 214, 220 application – attribute value – for role on symbol, 137 application/omdoc+xml 38 applied – attribute value – for role on presentation, 194, 330 apply – element, 256 m:apply – element, 17, 18, 115, 137, 194, 198, 238 archiving – document, 35 argument – element, 56, 156, 157, 320, 328 arith1 28, 219 arith1.ocd 20 artefacts – electronic, 102 artificial – intelligence, 286 assertion 69, 159, 162, 304 – level, 309 assertion – element, 69, 70, 77, 139, 141, 142, 145–148, 161, 164, 165, 178, 184, 229, 283, 314, 316, 320, 328, 340, 343 assertion – attribute
kohlhase-omdoc_PIC_0001	33	—	—	—	$\langle \text{FMP} \rangle \forall a, b. a + b = b + a \langle / \text{FMP} \rangle$
kohlhase-omdoc_PIC_0002	41	—	—	—	
kohlhase-omdoc_PIC_0003	41	—	—	—	

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kohlhase-omdoc_PIC_0004	68	—	—	—	$\langle \text{FMP} \rangle \forall x. 0 \leq x \wedge \forall x, y. x \leq y \Rightarrow s(x) \leq s(y) \langle / \text{FMP} \rangle$
kohlhase-omdoc_PIC_0005	72	—	—	—	$\langle \text{axiom xml:id="unit.ax"} \rangle \langle \text{FMP} \rangle \exists x \in M. \forall y \in M. y \circ x = y \langle / \text{FMP} \rangle \langle / \text{axiom} \rangle$
kohlhase-omdoc_PIC_0006	72	—	—	—	$\langle \text{assertion xml:id="unit.unique"} \rangle \langle \text{FMP} \rangle \exists^1 x \in M. \forall y \in M. y \circ x = y \langle / \text{FMP} \rangle \langle / \text{assertion} \rangle$
kohlhase-omdoc_PIC_0007	72	—	—	—	$\langle \text{assertion xml:id="left.unit"} \rangle \langle \text{FMP} \rangle \forall x \in M. e \circ x = x \langle / \text{FMP} \rangle \langle / \text{assertion} \rangle$
kohlhase-omdoc_PIC_0008	72	—	—	—	$\forall S \subseteq M. S^* = S \setminus \{e\}$
kohlhase-omdoc_PIC_0009	73	—	—	—	$\langle \text{axiom xml:id="inv.ax"} \rangle \langle \text{FMP} \rangle \forall x \in M. \exists y \in M. x \circ y = e \langle / \text{FMP} \rangle \langle / \text{axiom} \rangle$
kohlhase-omdoc_PIC_0010	93	—	—	—	<i>Formulae are a central part of mathematical documents; this module integrates the content-oriented representation formats OPENMATH and MATHML into OMDoc</i>
kohlhase-omdoc_PIC_0011	93	—	—	—	<i>Markup for mathematical forms like theorems, axioms, definitions, and examples that can be used to specify or define properties of given mathematical objects and theories to group mathematical statements and provide a notion of context.</i>
kohlhase-omdoc_PIC_0012	93	—	—	—	<i>Structure of proofs and argumentations at various levels of details and formality</i>
kohlhase-omdoc_PIC_0013	131	—	—	—	$\langle \text{FMP logic="fol"} \rangle \forall V, F. \overline{\text{binop}}(F, V) \Leftrightarrow \mathbf{Im}(F) = V \wedge \mathbf{Dom}(F) = V \langle / \text{FMP} \rangle$
kohlhase-omdoc_PIC_0014	131	—	—	—	$\langle \text{CMP} \rangle \text{If } a \in U \text{ and } a \in V, \text{ then } a \in U \cap V \text{ or}$
kohlhase-omdoc_PIC_0015	148	—	—	—	$\langle \text{FMP} \rangle \forall S. \text{sgrp}(S) \rightarrow \forall x, y. \text{unit}(x, S) \wedge \text{unit}(y, S) \rightarrow x = y \langle / \text{FMP} \rangle$
kohlhase-omdoc_PIC_0016	158	—	—	—	$\langle \text{CMP} \rangle \text{neut is a left unit for op.} \langle / \text{CMP} \rangle \langle \text{FMP} \rangle \forall x \in \mathbf{set}. \text{op}(x, \text{neut}) = x \langle / \text{FMP} \rangle$
kohlhase-omdoc_PIC_0017	158	—	—	—	$\langle \text{CMP} \rangle \cdot^* \text{ subtracts the unit from a set} \langle / \text{CMP} \rangle \langle \text{FMP} \rangle \forall S. S^* = S \setminus \{\mathbf{unit}\} \langle / \text{FMP} \rangle$
kohlhase-omdoc_PIC_0018	167	—	—	—	$\langle \text{CMP} \rangle \text{Let } q \stackrel{def}{=} p_1 \cdots p_n + 1 \langle / \text{CMP} \rangle$
kohlhase-omdoc_PIC_0019	172	—	—	—	$\langle \text{CMP} \rangle \text{If } a \in U \text{ or } a \in V, \text{ then } a \in U \cup V. \langle / \text{CMP} \rangle$
kohlhase-omdoc_PIC_0020	172	—	—	—	$\langle \text{CMP} \rangle \text{If } a \in U, \text{ then } a \in U \cup V. \langle / \text{CMP} \rangle$
kohlhase-omdoc_PIC_0021	172	—	—	—	$\langle \text{CMP} \rangle \text{If } a \in V, \text{ then } a \in U \cup V. \langle / \text{CMP} \rangle$
kohlhase-omdoc_PIC_0022	173	—	—	—	$\langle \text{assumption xml:id="t1_p1_l1.d1.a"} \rangle a \in U \langle / \text{assumption} \rangle$
kohlhase-omdoc_PIC_0023	174	—	—	—	$\langle \text{derive xml:id="d3"} \rangle \langle \text{CMP} \rangle \text{Then } P = \{p_1, \dots, p_n\} \text{ for some } n \langle / \text{CMP} \rangle \langle / \text{derive} \rangle$
kohlhase-omdoc_PIC_0024	175	—	—	—	$\langle \text{CMP} \rangle \text{Let } q \stackrel{def}{=} p_1 \cdots p_n + 1 \langle / \text{CMP} \rangle$
kohlhase-omdoc_PIC_0025	184	—	—	—	$\langle \text{assertion xml:id="conv.inv"} \text{ theory="group"} \rangle \forall x, y \in M. x \circ x^{-1} = e \langle / \text{assertion} \rangle$
kohlhase-omdoc_PIC_0026	246	—	—	—	Symbol: Σ APPLICATION "arith1:sum"